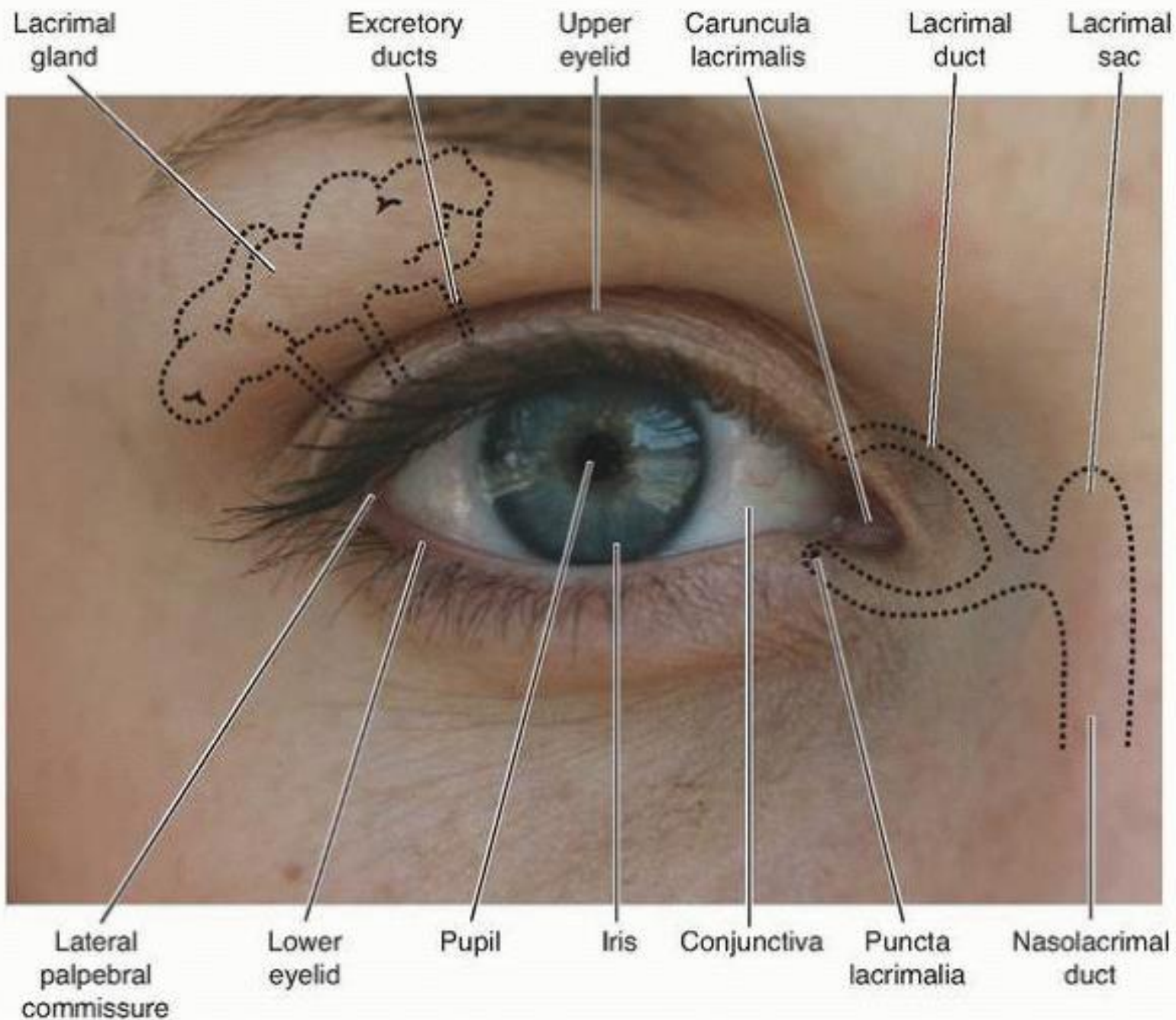
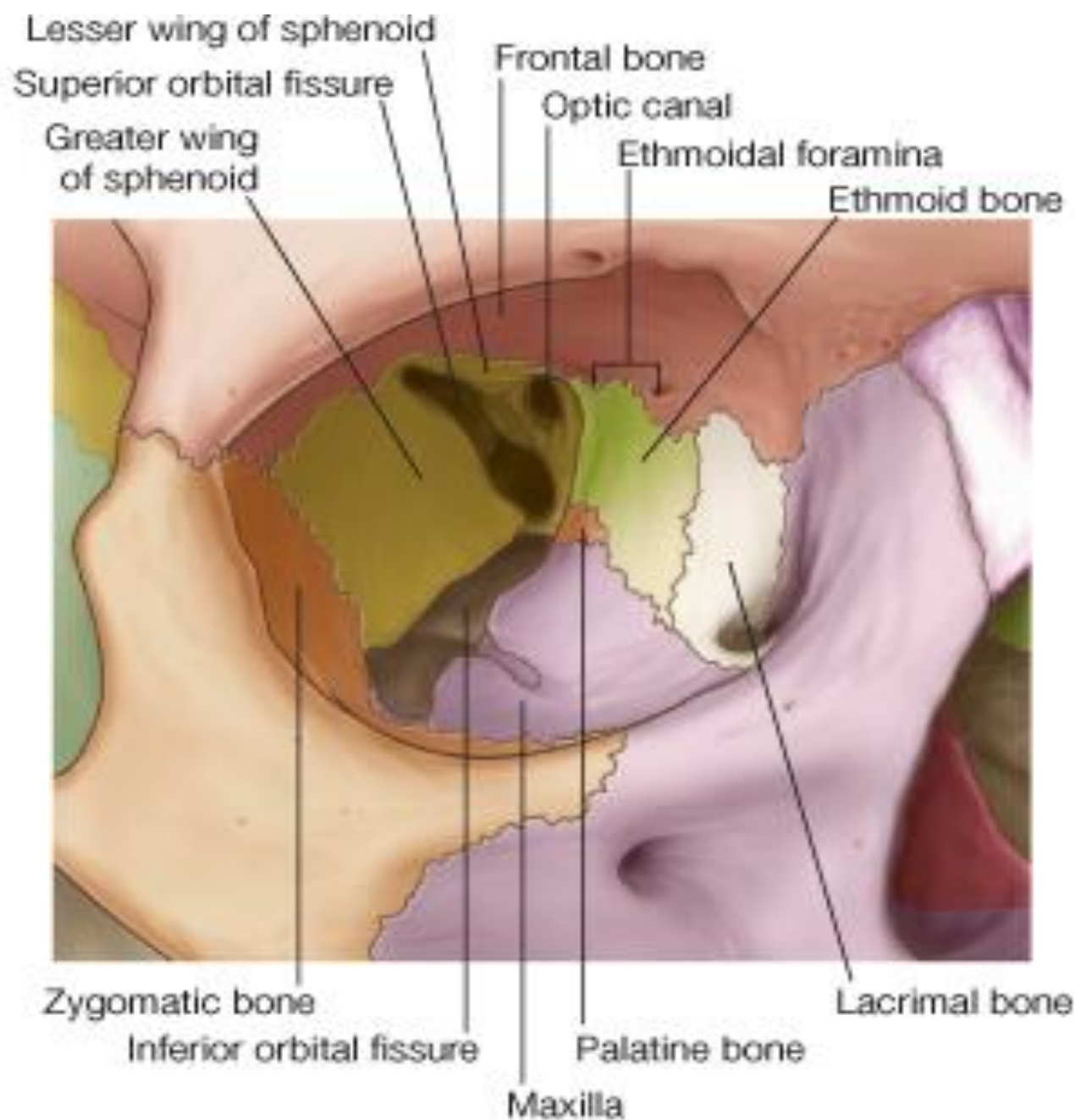


ORBITAL REGION, ORBIT, EYE, AND EYEBALL



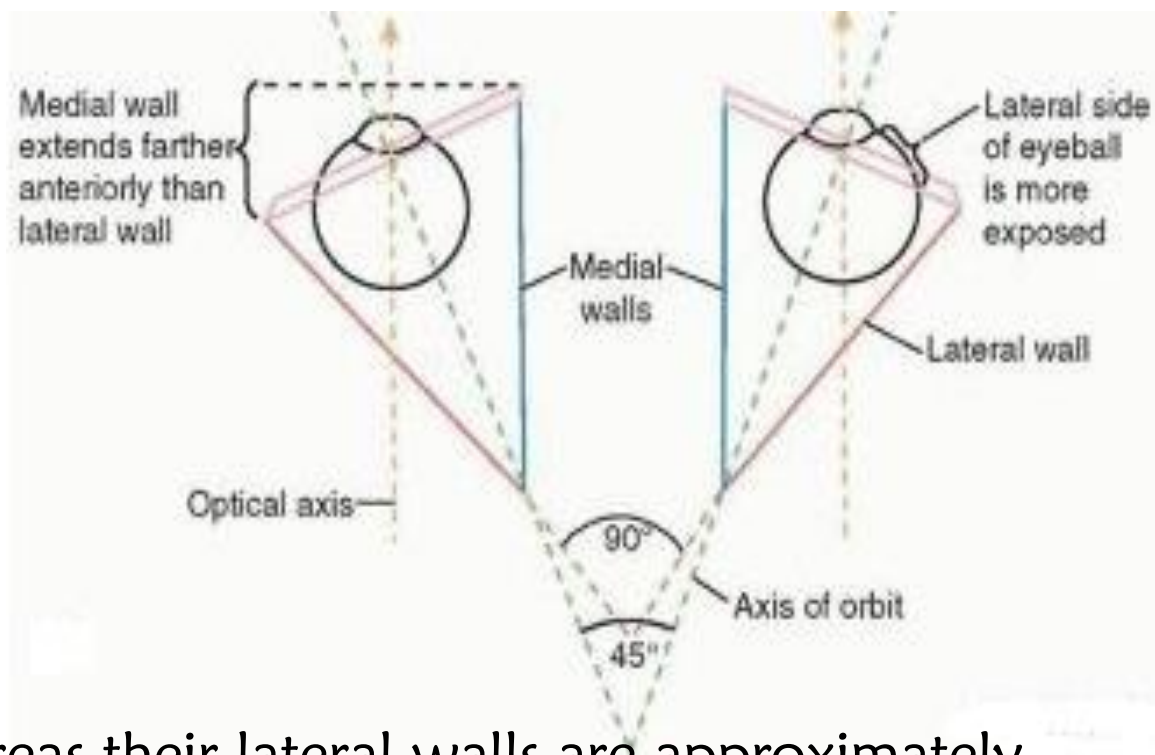
Orbits

- is a holloquadrangular pyramidal cavity with its base in front (anterolaterally) and its apex behind (posteromedially)
- The orbital margin is formed:
 - superiorly margin by the frontal bone
 - the lateral margin is formed by the processes of the frontal and zygomatic bones
 - the inferior margin is formed by the zygomatic bone and the maxilla
 - the medial margin is formed by the processes of the maxilla and the frontal bone.



The medial walls of the two orbits:

- separated by the ethmoidal sinuses and the upper parts of the nasal cavity



- are nearly parallel, whereas their lateral walls are approximately at a right (90°) angle.
- the axes of the orbits (**orbital axes**) diverge at approximately 45°.
- The optical axes** (axes of gaze, the direction or line of sight) for the two eyeballs, are parallel,
- however, and in the anatomical position run directly anteriorly (“looking straight ahead”), the eyeballs being in the primary position

The quadrangular pyramidal orbit has:

- a base
- four walls, and an apex

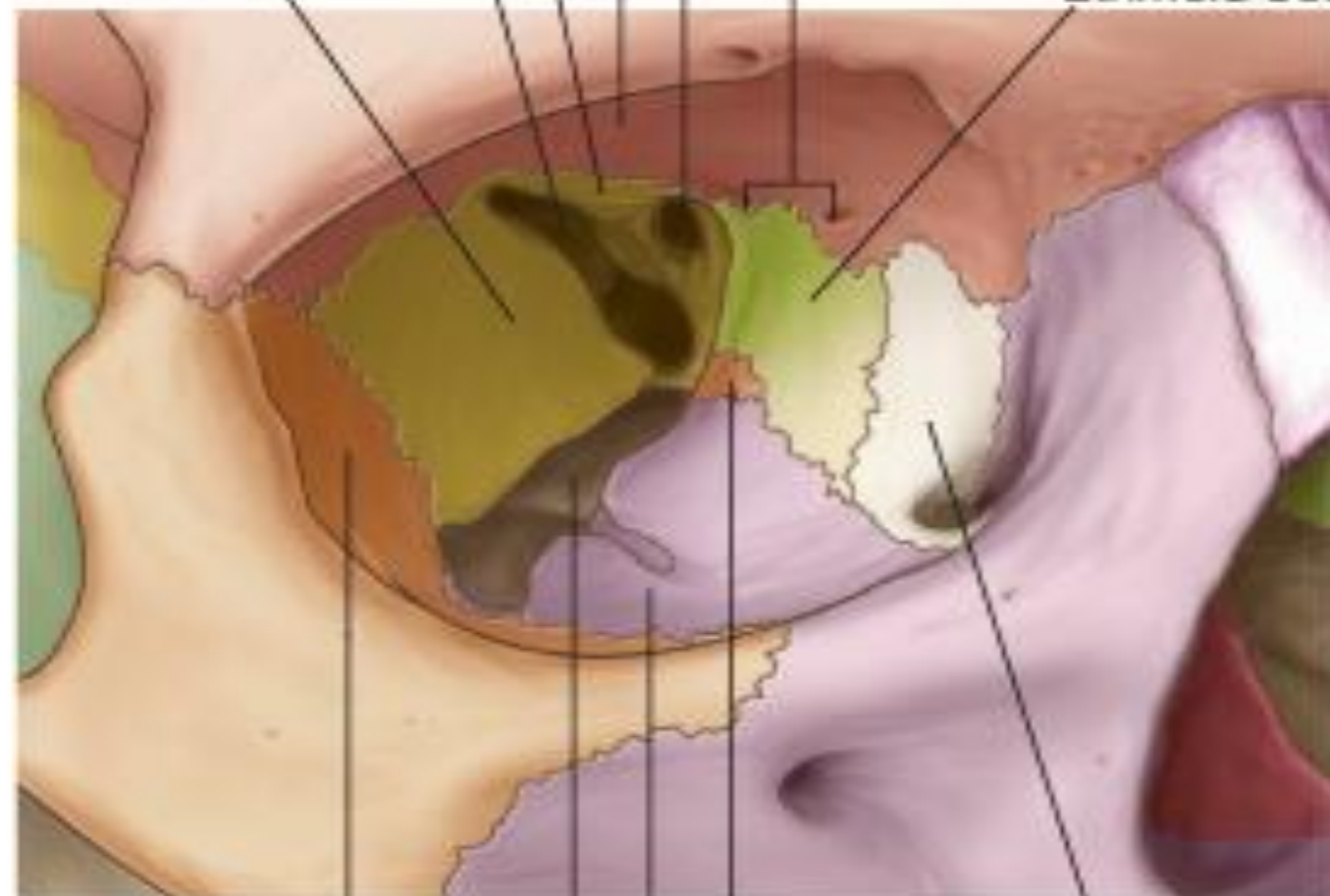
- **The base of the orbit:**

- is outlined by the orbital margin, which surrounds the orbital opening.
- The bone forming the orbital margin is reinforced to afford protection to the orbital contents and provides attachment for the orbital septum, a fibrous membrane that extends into the eyelids.

- **The superior wall (roof):**

- is approximately horizontal and is formed mainly by the orbital part of the frontal bone, which separates the orbital cavity from the anterior cranial fossa.
- Near the apex of the orbit, the superior wall is formed by the lesser wing of the sphenoid.
- **Anterolaterally**, a shallow depression in the orbital part of the frontal bone, called the **fossa for the lacrimal gland** (lacrimal fossa), accommodates **the lacrimal gland**

Lesser wing of sphenoid
Superior orbital fissure
Greater wing of sphenoid
Frontal bone
Optic canal
Ethmoidal foramina
Ethmoid bone

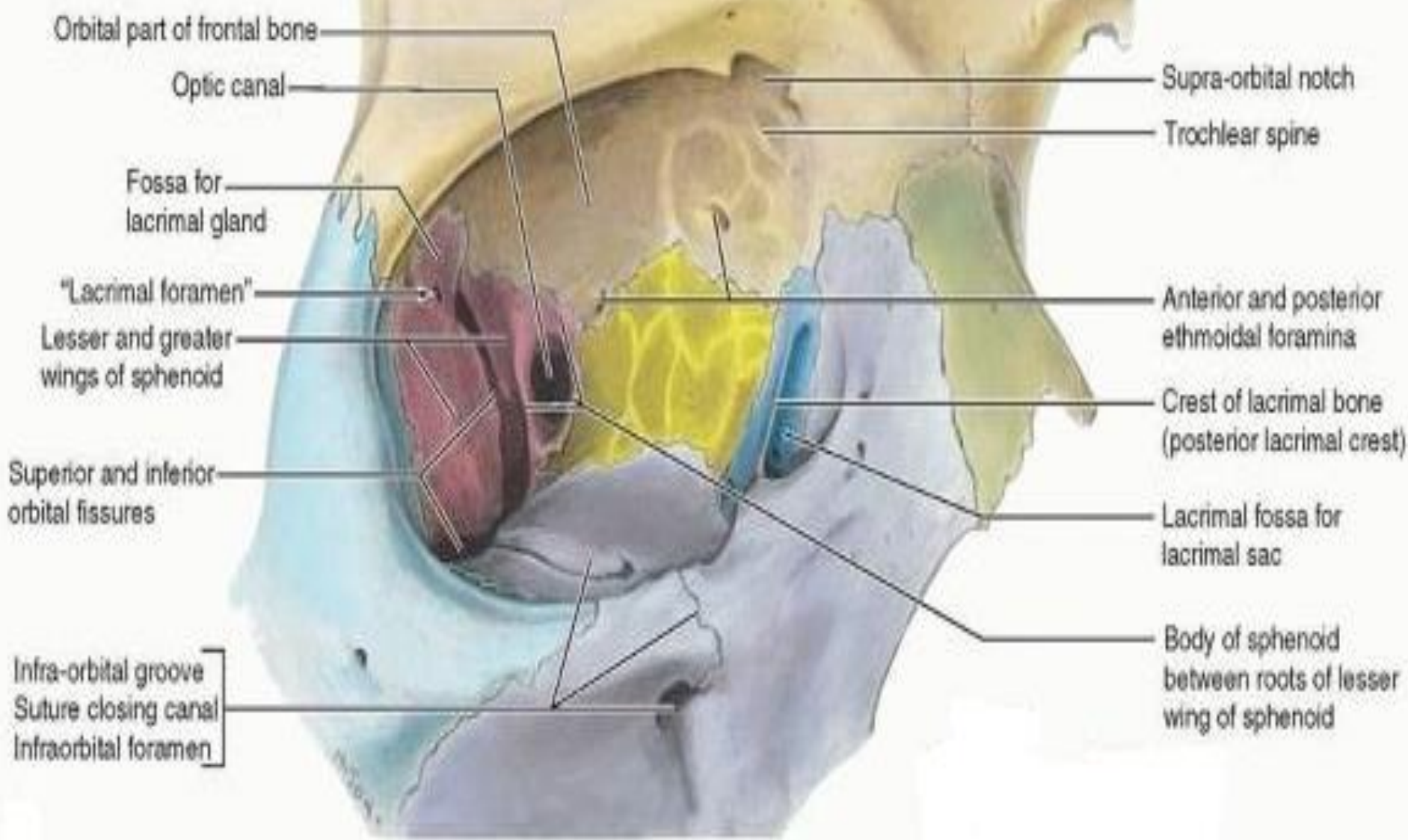


Zygomatic bone
Inferior orbital fissure
Palatine bone
Maxilla
Lacrimal bone

- **The medial walls:**
 - the contralateral orbits are essentially parallel
 - and are formed primarily by
 - the orbital plate of the ethmoid bone, along with contributions from the frontal process of the maxilla, lacrimal, and sphenoid bones.
 - **Anteriorly**, the medial wall is indented by the lacrimal groove and fossa for the lacrimal sac
 - the trochlea (pulley) for the tendon of one of the extraocular muscles is located superiorly
 - Much of the bone forming the medial wall is paper thin
 - the ethmoid bone is highly pneumatized with ethmoidal cells, often visible through the bone of a dried cranium

- **The inferior wall (orbital floor):**
 - is formed mainly by the maxilla and partly by the zygomatic and palatine bones.
 - The thin inferior wall is shared by the orbit and maxillary sinus.
 - It slants inferiorly from the apex to the inferior orbital margin
 - The inferior wall is demarcated from the lateral wall of the orbit by the inferior orbital fissure, a gap between the orbital surfaces of the maxilla and the sphenoid.

- **The lateral wall:**
 - is formed by the frontal process of the zygomatic bone and the greater wing of the sphenoid.
 - This is **the strongest and thickest wall**, which is important because it is most exposed and vulnerable to direct trauma.
 - Its posterior part separates the orbit from the temporal and middle cranial fossae.
 - The lateral walls of the contralateral orbits are nearly **perpendicular to each other**
- **The apex of the orbit:**
 - is at the optic canal in the lesser wing of the sphenoid just medial to the superior orbital fissure



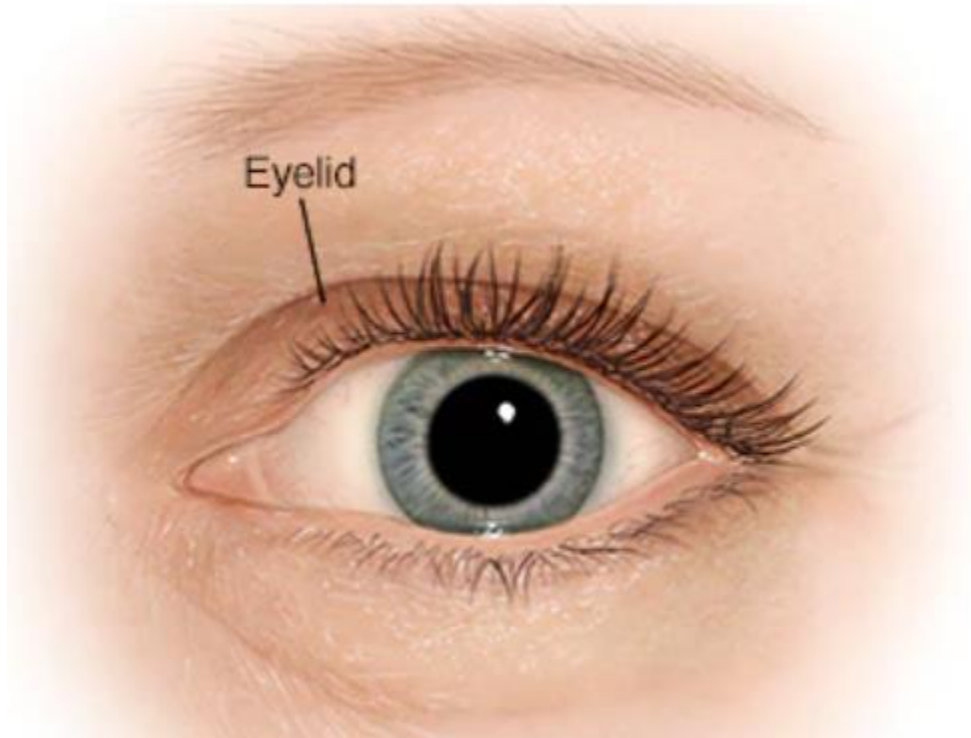
Anterolateral view

- The bones forming the orbit:
 - are lined with periorbita, the periosteum of orbit.
 - **The periorbita is continuous:**
 - at the optic canal and superior orbital fissure with the **periosteal layer of the dura mater**.
 - over the orbital margins and through the inferior orbital fissure with the periosteum covering the **external surface the cranium**
 - with the orbital septa at the orbital margins.
 - with the fascial sheaths of the extraocular muscles.
 - with the orbital fascia that forms the fascial sheath of the eyeball.

- **Contents of the orbits and orbital region:**
 - **Eyelids**, which bound the orbits anteriorly, controlling exposure of the anterior eyeball.
 - **Lacrimal** apparatus
 - **Extraocular muscles**, which position the eyeballs and raise the superior eyelids.
 - **Nerves and vessels** in transit to the eyeballs and muscles.
 - **Orbital fascia** surrounding the eyeballs and muscles.
 - **Mucous membrane** (conjunctiva) lining the eyelids and anterior aspect of the eyeballs and most of the lacrimal apparatus, which lubricates it.

1. Eyelids and Lacrimal Apparatus

- The eyelids and lacrimal fluid, secreted by the lacrimal glands
- protect the cornea and eyeballs from injury and irritation (e.g., by dust and small particles).



EYELIDS

- are moveable folds that cover the eyeball anteriorly
- keep the cornea moist by spreading the lacrimal fluid.
- The eyelids are covered externally by **thin skin** and internally by transparent mucous membrane, **the palpebral conjunctiva**
- This part of the conjunctiva is reflected onto the eyeball, where it is continuous with the **bulbar conjunctiva**.
- The lines of reflection of the **palpebral conjunctiva** onto the eyeball form deep recesses, the **superior and inferior conjunctival fornices**

- **The conjunctival sac:**

- is the space bound by the palpebral and bulbar conjunctivae
- it is a closed space when the eyelids are closed, but opens via an anterior aperture, the palpebral fissure when the eye is open.
- is a specialized form of mucosal “bursa” that enables the eyelids to move freely over the surface of the eyeball as they open and close.

- **The superior (upper) and inferior (lower) eyelids:**

- are strengthened by dense bands of connective tissue, the superior and inferior tarsi (sing. = tarsus) which form the “**skeleton**” of the eyelids

- Embedded in the **tarsi are tarsal glands**,
 - the lipid secretion of which lubricates the edges of the eyelids
 - prevents them from sticking together when they close.
- The lipid secretion also forms a barrier that **lacrimal fluid does not cross when produced in normal amounts**.
- When production is excessive, it spills over the barrier onto the cheeks as tears
- These glands, along with the **ciliary glands** (modified sudoriferous glands located in the margin), open via small pores onto the margin adjacent to the eyelashes

Levator palpebrae
superioris muscle

Periosteum

Orbicularis oculi
muscle

Orbital septum

Tendon of
levator palpebrae
superioris muscle

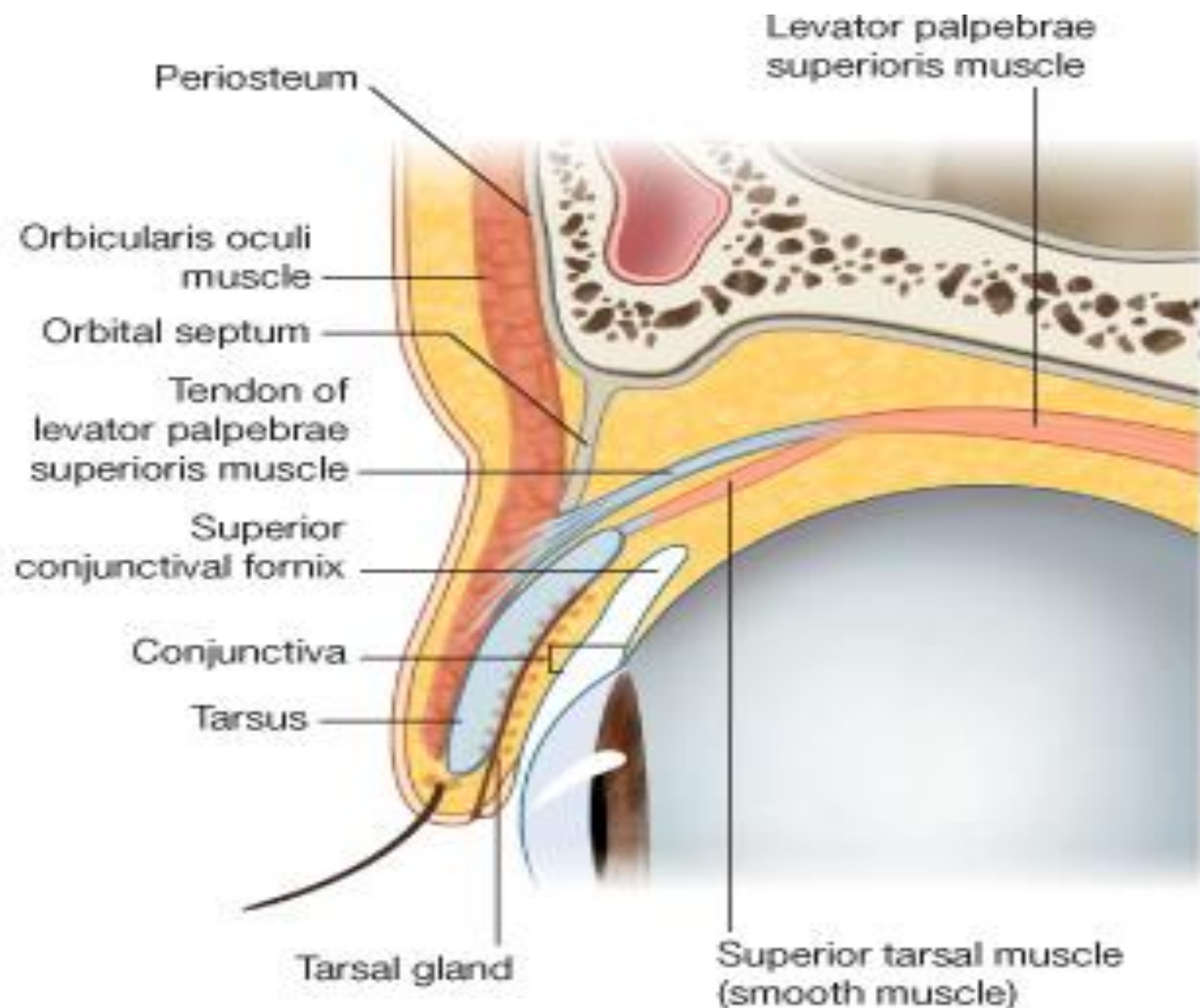
Superior
conjunctival fornix

Conjunctiva

Tarsus

Tarsal gland

Superior tarsal muscle
(smooth muscle)



- **The eyelashes:**
 - are in the margins of the lids.
 - The large sebaceous glands associated with the eyelashes are **ciliary glands**.
- The junctions of the **superior** and **inferior eyelids** make up the medial and lateral **palpebral commissures**, defining the medial and lateral angles of the eye
- Between the nose and the medial angle of the eye is the **medial palpebral ligament**, which connects the tarsi to the medial margin of the orbit
- The orbicularis oculi originates and inserts onto this ligament.
- A similar **lateral palpebral ligament** attaches the tarsi to the lateral margin of the orbit but does not provide for direct muscle attachment

Tendon of levator
palpebrae superioris muscle

Orbital septum

Anterior lacrimal crest

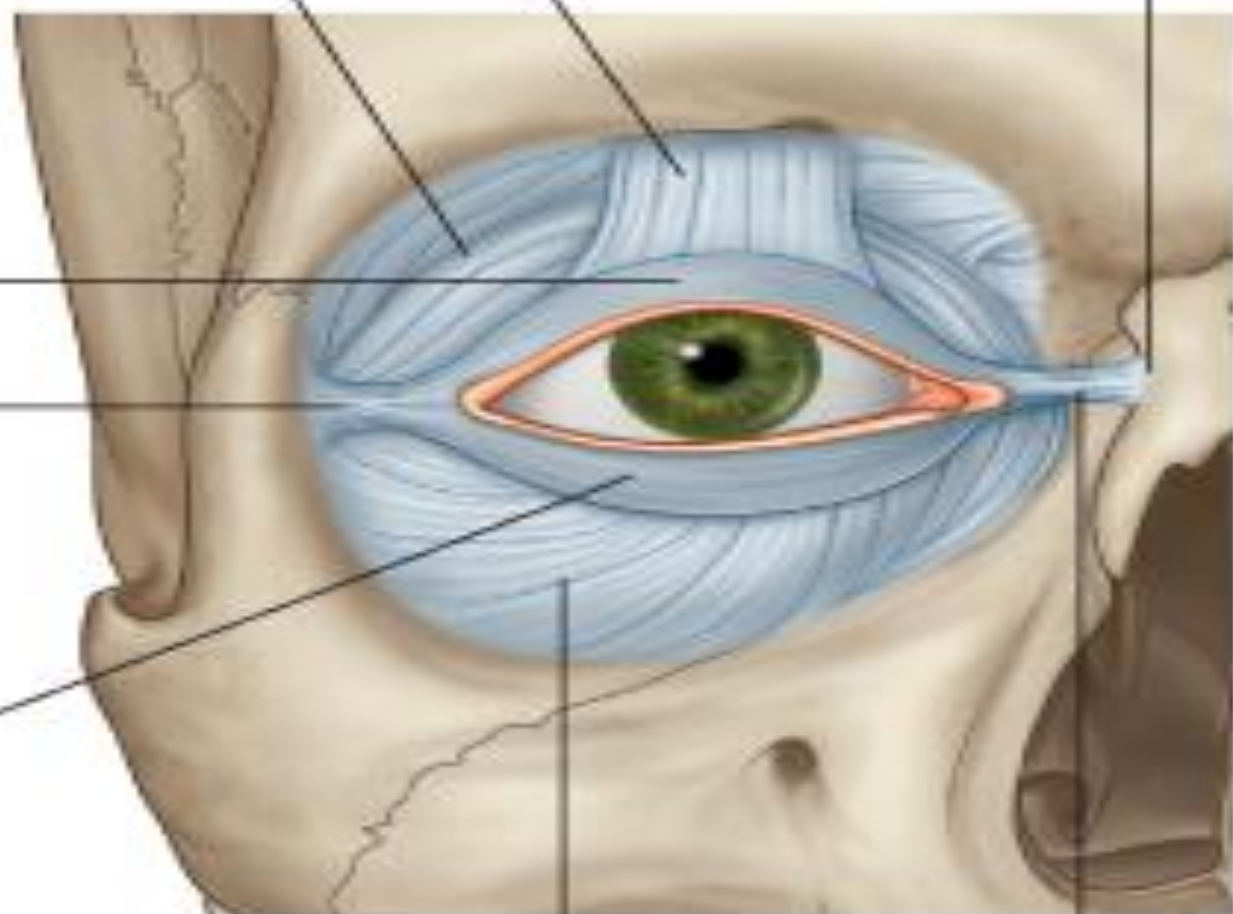
Superior tarsus

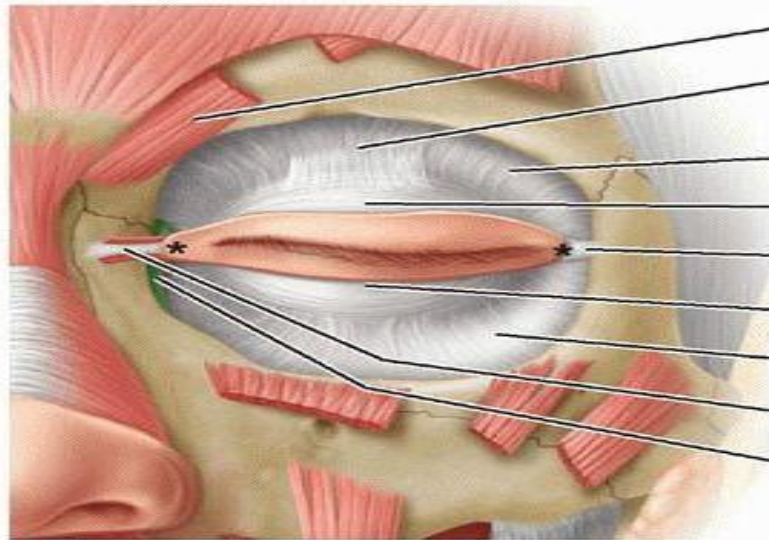
Lateral palpebral
ligament

Inferior tarsus

Orbital septum

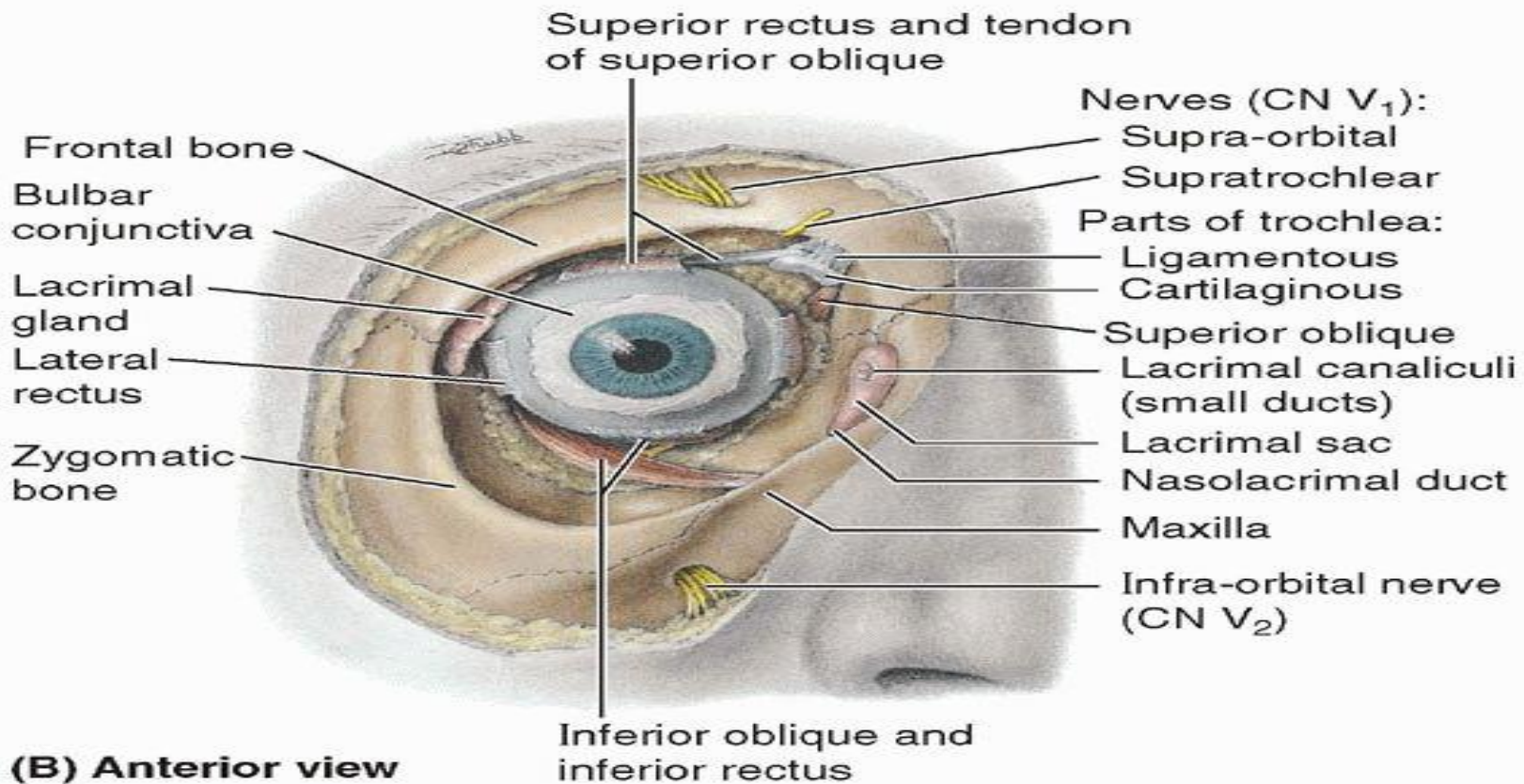
Medial palpebral ligament





- Corrugator supercilii muscle (cut)
- Tendon of levator palpebrae superioris muscle
- Superior orbital septum
- Superior tarsus
- Lateral palpebral ligament
- Inferior tarsus
- Inferior orbital septum
- Medial palpebral ligament
- Lacrimal sac
- * Medial and lateral palpebral commissures

(A) Anterior view

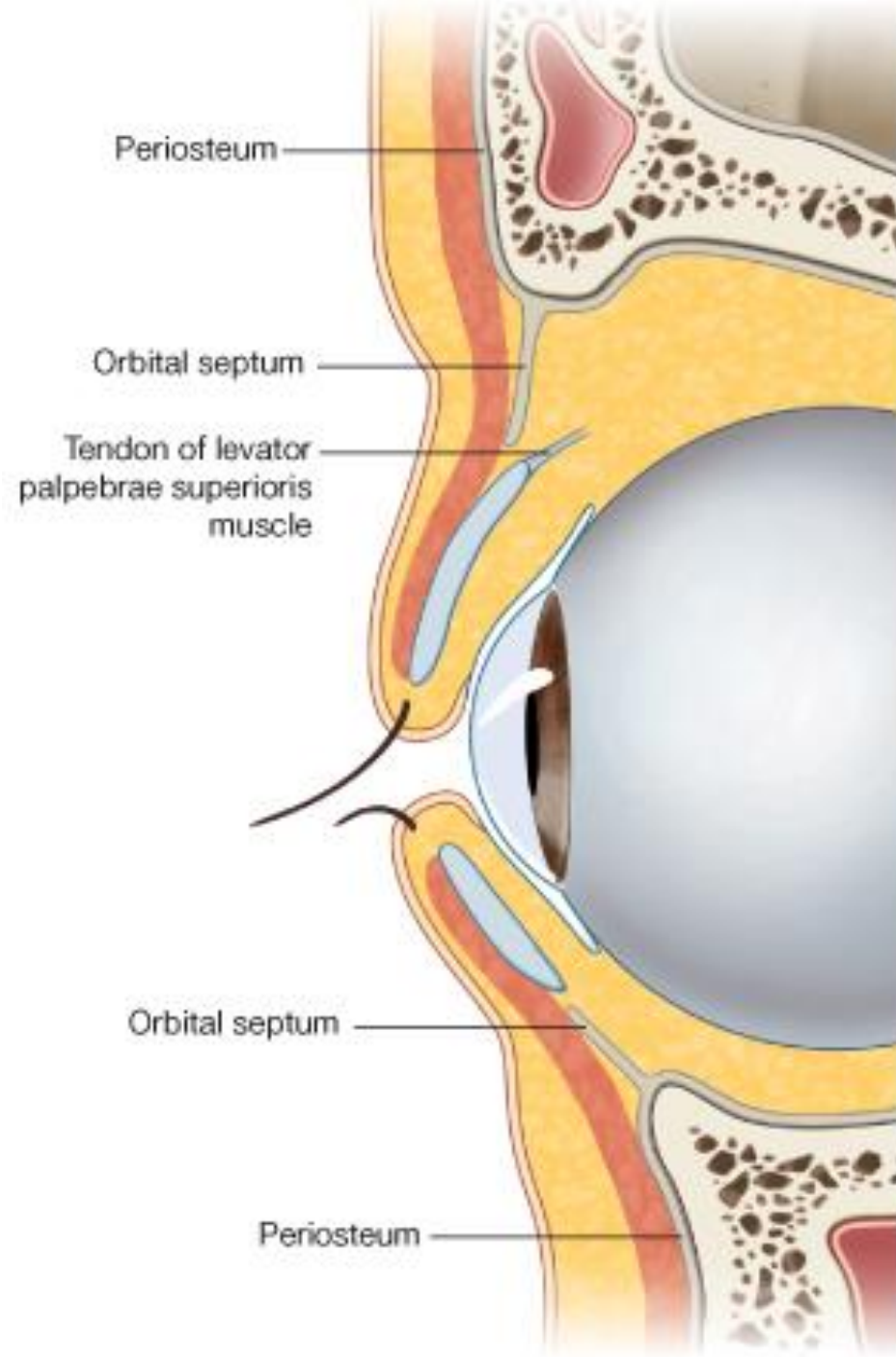


(B) Anterior view

- **The orbital septum:**

- is a fibrous membrane that spans from the tarsi to the margins of the orbit, where it becomes continuous with the periosteum
- It keeps the orbital fat contained and,
- owing to its continuity with the **periorbital**, can limit the spread of infection to and from the orbit.

The septum constitutes in large part the posterior fascia of the orbicularis oculi muscle.



LACRIMAL APPARATUS

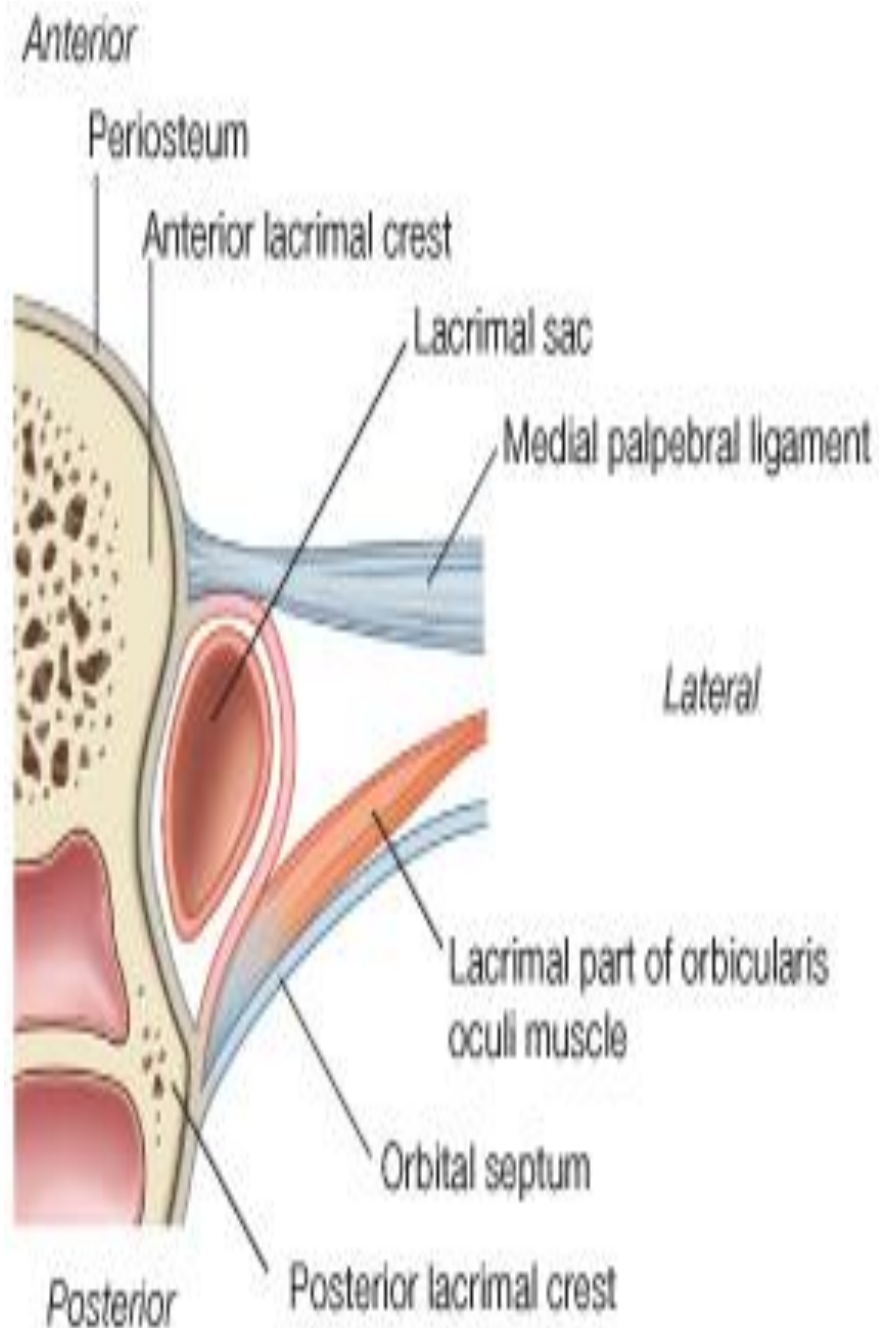
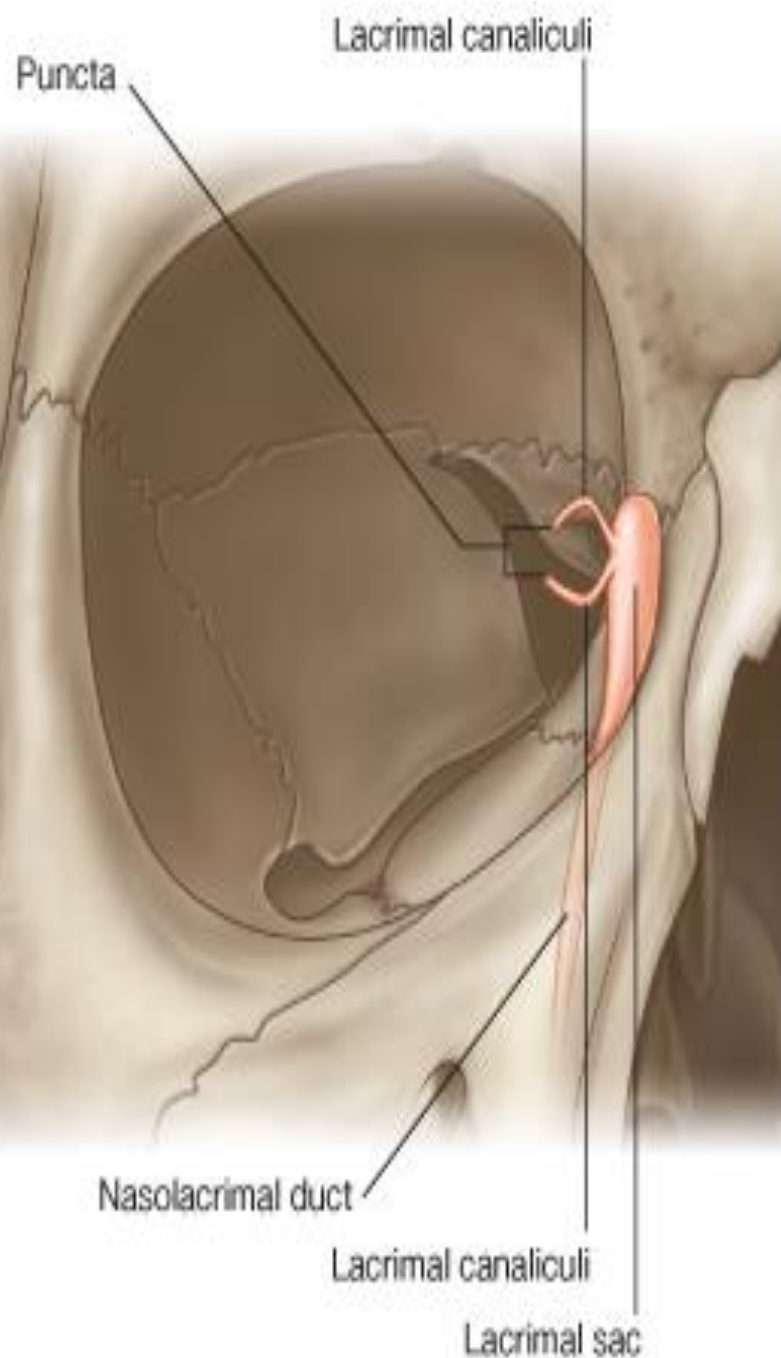
- The lacrimal apparatus consists of the:
 - **Lacrimal gland:**
 - secretes lacrimal fluid, a watery physiological saline containing the bacteriocidal enzyme lysozyme
 - and provides some nutrients and dissolved oxygen to the cornea
 - **Excretory ducts:**
 - convey lacrimal fluid from the lacrimal glands to the conjunctival sac

– Lacrimal canaliculi:

- commence at a **lacrimal punctum** (opening) on the lacrimal papilla near the medial angle of the eye and drain lacrimal fluid from the lacrimal lake

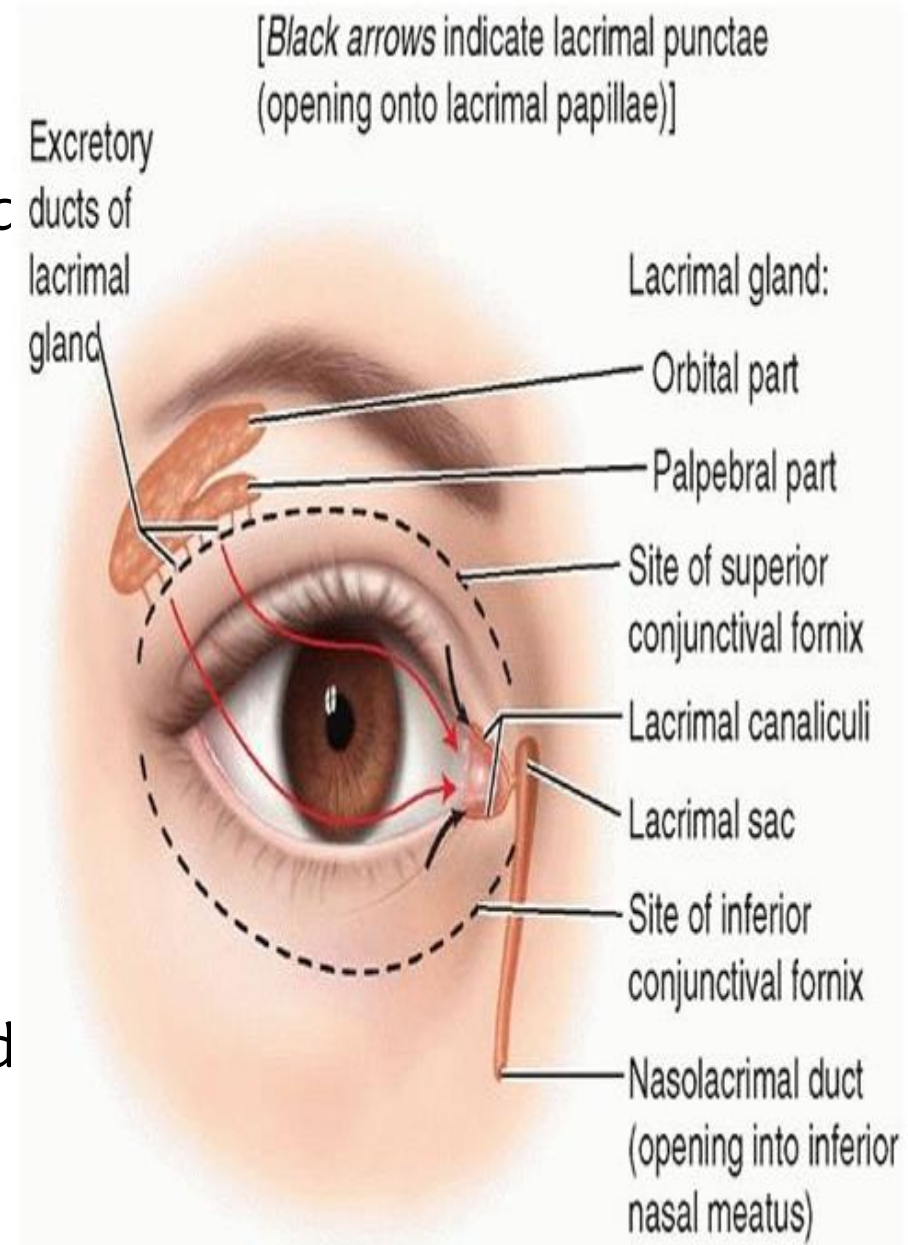
– lacrimal sac:

- a triangular space at the medial angle of the eye where the tears collect
- is the dilated superior part of the nasolacrimal duct



Nasolacrimal duct:

- is about 0.5 in. (1.3 cm) long and emerges from the lower end of the lacrimal sac
- The duct descends downward, backward, and laterally in a bony canal and opens into the inferior meatus of the nose.
- The opening is guarded by a fold of mucous membrane known **as the lacrimal fold**.



[Black arrows indicate lacrimal punctae (opening onto lacrimal papillae)]

Excretory ducts of lacrimal gland

Lacrimal gland:

Orbital part

Palpebral part

Site of superior conjunctival fornix

Lacrimal canaliculi

Lacrimal sac

Site of inferior conjunctival fornix

Nasolacrimal duct (opening into inferior nasal meatus)

Lateral angle (canthus)

Inferior conjunctival fornix

Plica semilunaris of conjunctiva

Lacrimal caruncle in lacrimal lake

Inferior lacrimal papilla and punctum

Sclera

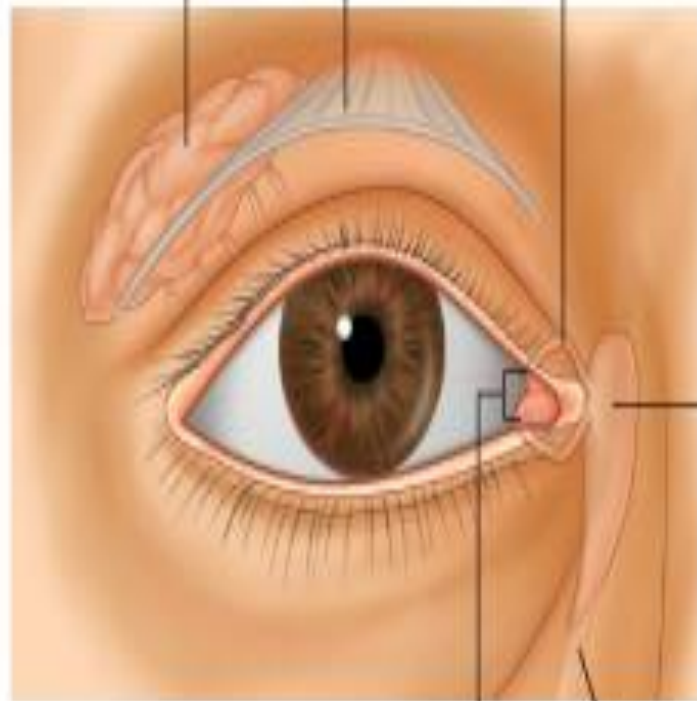
Conjunctival blood vessels

- **The lacrimal gland:**
 - almond shaped and approximately 2 cm long
 - lies in the fossa for the lacrimal gland in the **superolateral part of each orbit**
 - divided into superior (orbital) and inferior (palpebral) parts by the lateral expansion of the tendon of the levator palpebrae superioris
 - **Accessory lacrimal glands** are also present, sometimes in the middle part of the eyelid or along the superior or inferior fornices of the conjunctival sac.
 - They are more numerous in the superior eyelid than in the inferior eyelid

Tendon of levator palpebrae superioris muscle

Lacrimal gland

Lacrimal canaliculi



Medial

Lacrimal sac

Puncta

Nasolacrimal duct

[Black arrows indicate lacrimal punctae (opening onto lacrimal papillae)]

Excretory ducts of lacrimal gland

Lacrimal gland:

Orbital part

Palpebral part

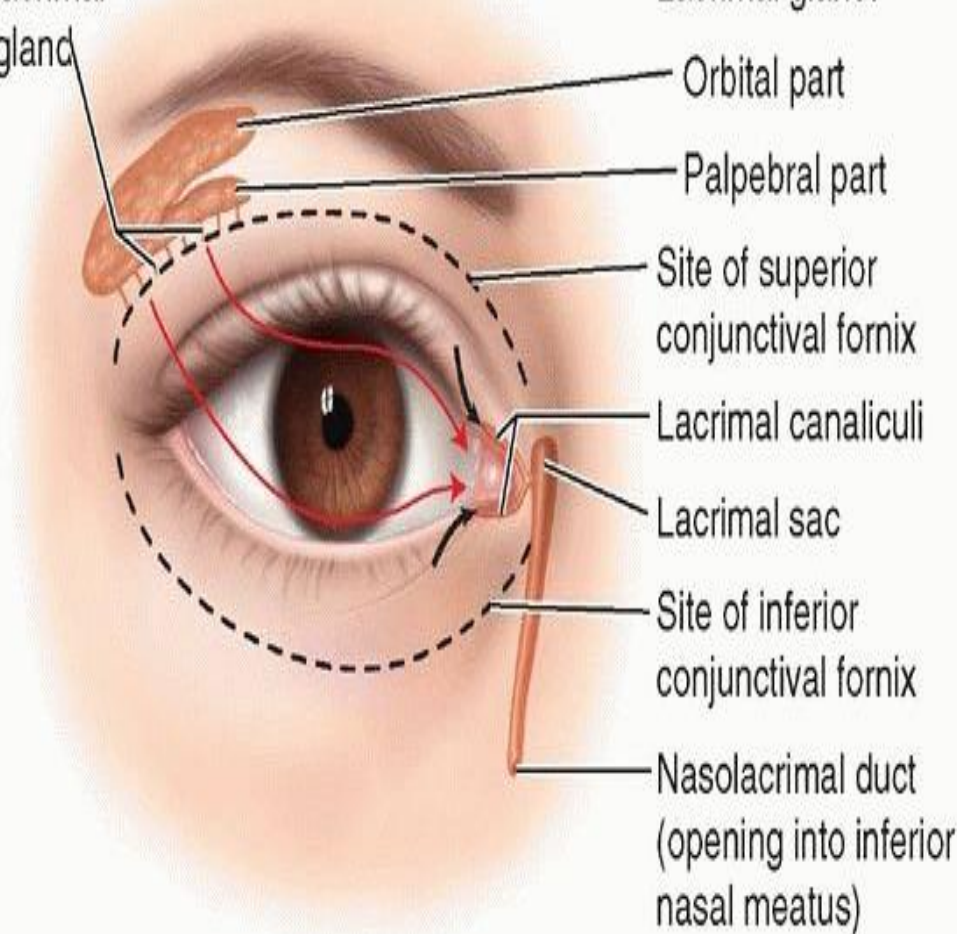
Site of superior conjunctival fornix

Lacrimal canaliculi

Lacrimal sac

Site of inferior conjunctival fornix

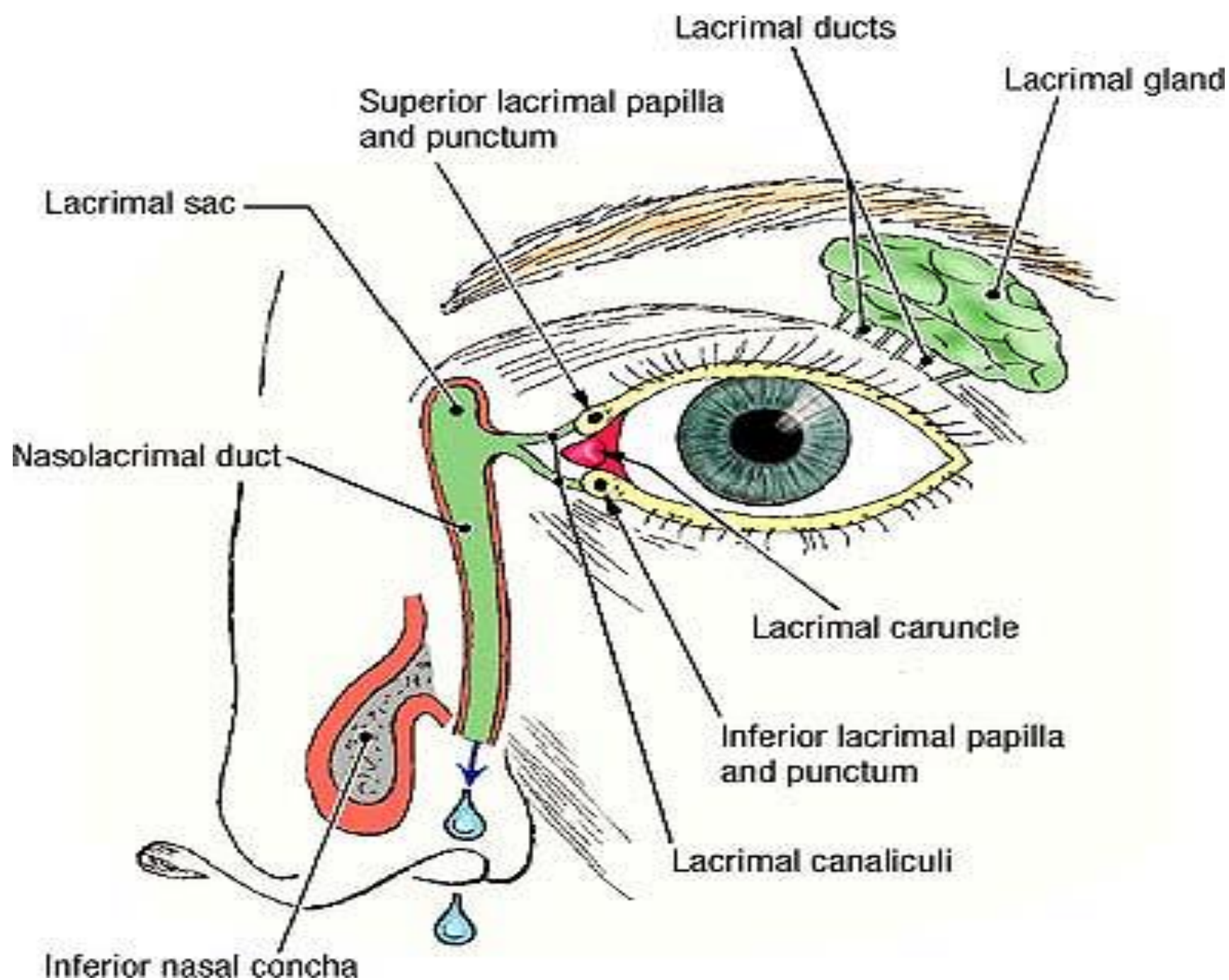
Nasolacrimal duct (opening into inferior nasal meatus)



Lacrimal gland Excretory ducts Upper eyelid Caruncula lacrimalis Lacrimal duct Lacrimal sac

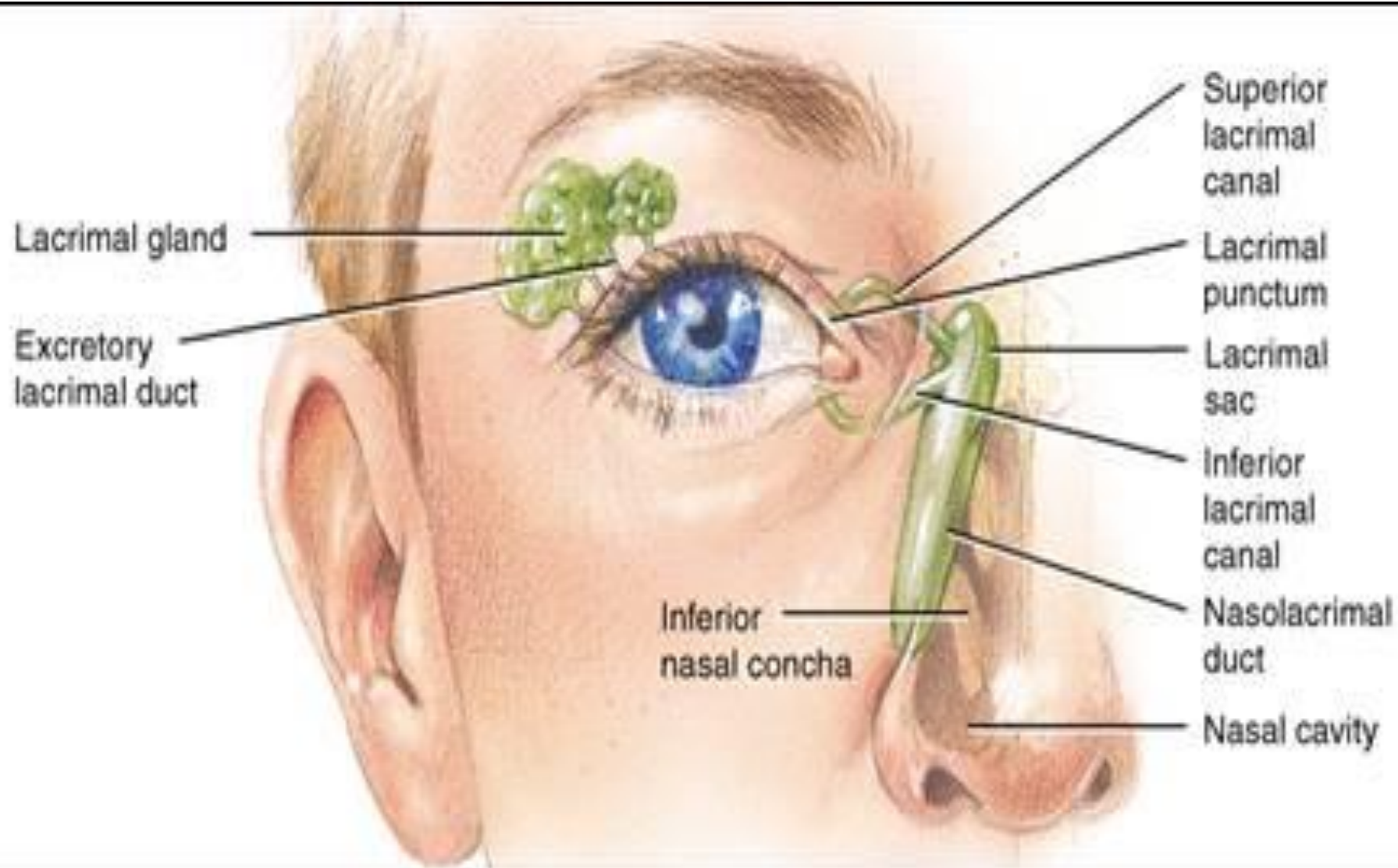


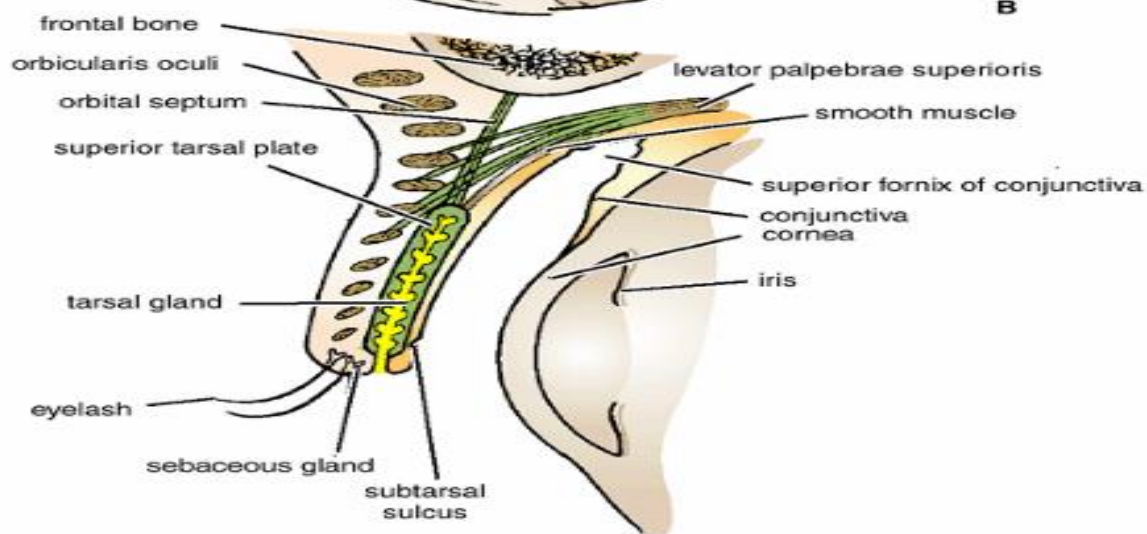
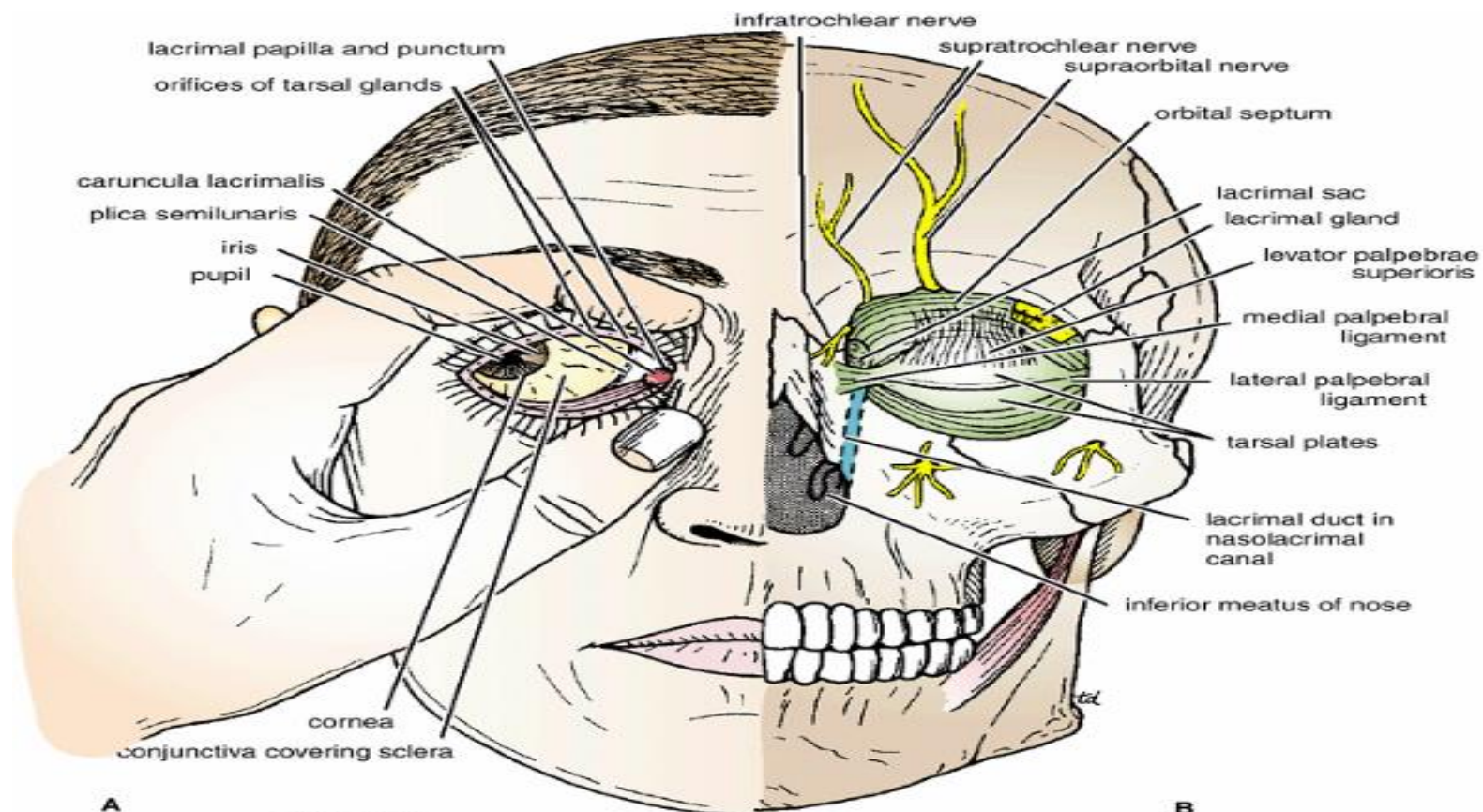
Lateral palpebral commissure Lower eyelid Pupil Iris Conjunctiva Puncta lacrimalia Nasolacrimal duct



Lacrimal Ducts

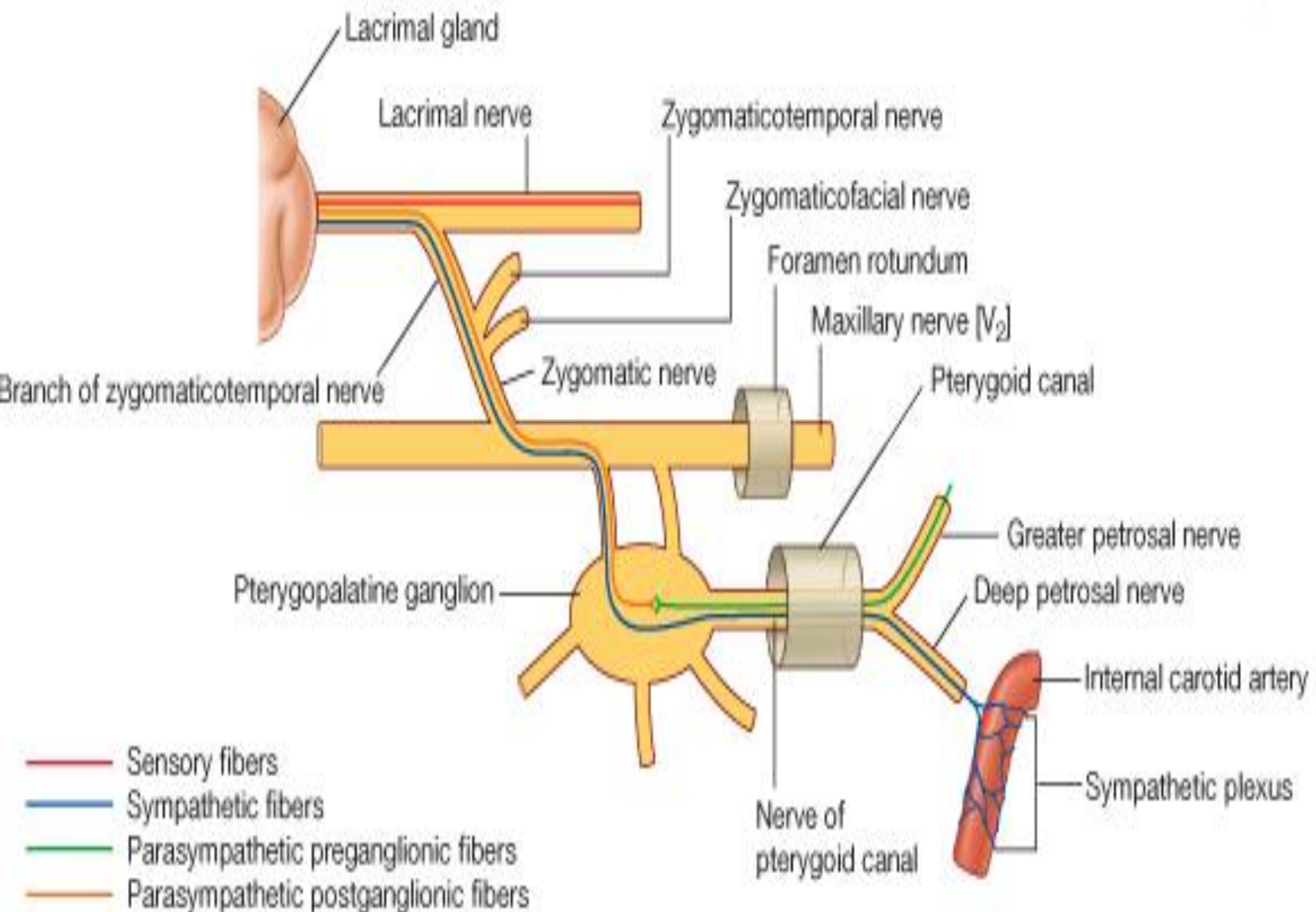
- The tears circulate across the cornea and accumulate in the lacus lacrimalis.
- From here, the tears enter the canaliculi lacrimales through the **puncta lacrimalis**.
- The canaliculi lacrimales pass medially and open into **the lacrimal sac**, which lies in the lacrimal groove behind the medial palpebral ligament and is the upper blind end of the nasolacrimal duct
- lacrimal fluid is stimulated by parasympathetic impulses from CN VII.
- It is secreted through **8-12 excretory ducts**, which open into the lateral part of the superior conjunctival fornix of the conjunctival sac





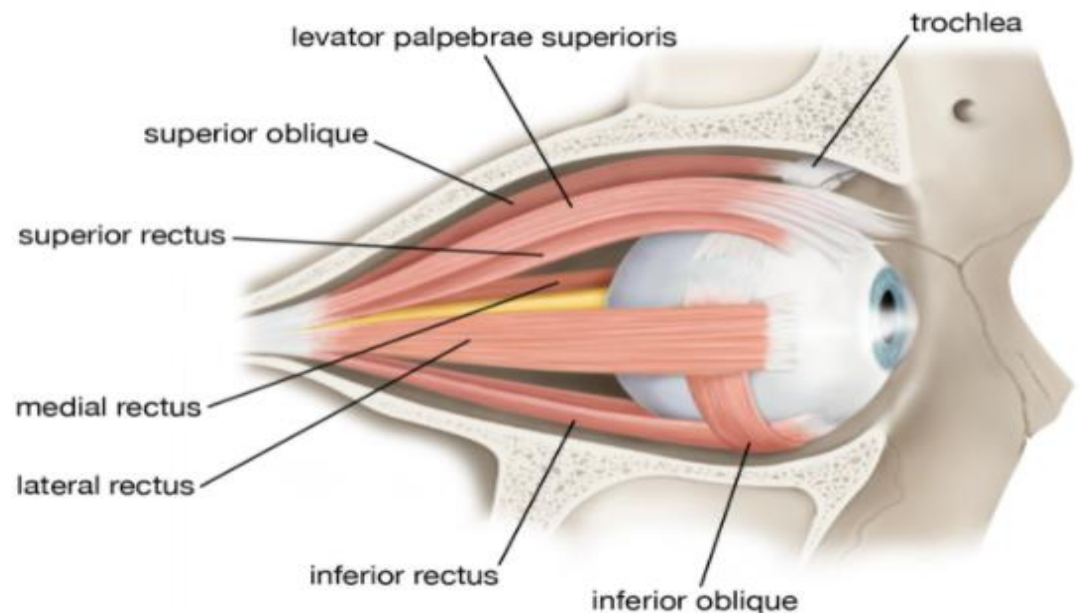
The nerve supply of the lacrimal gland:

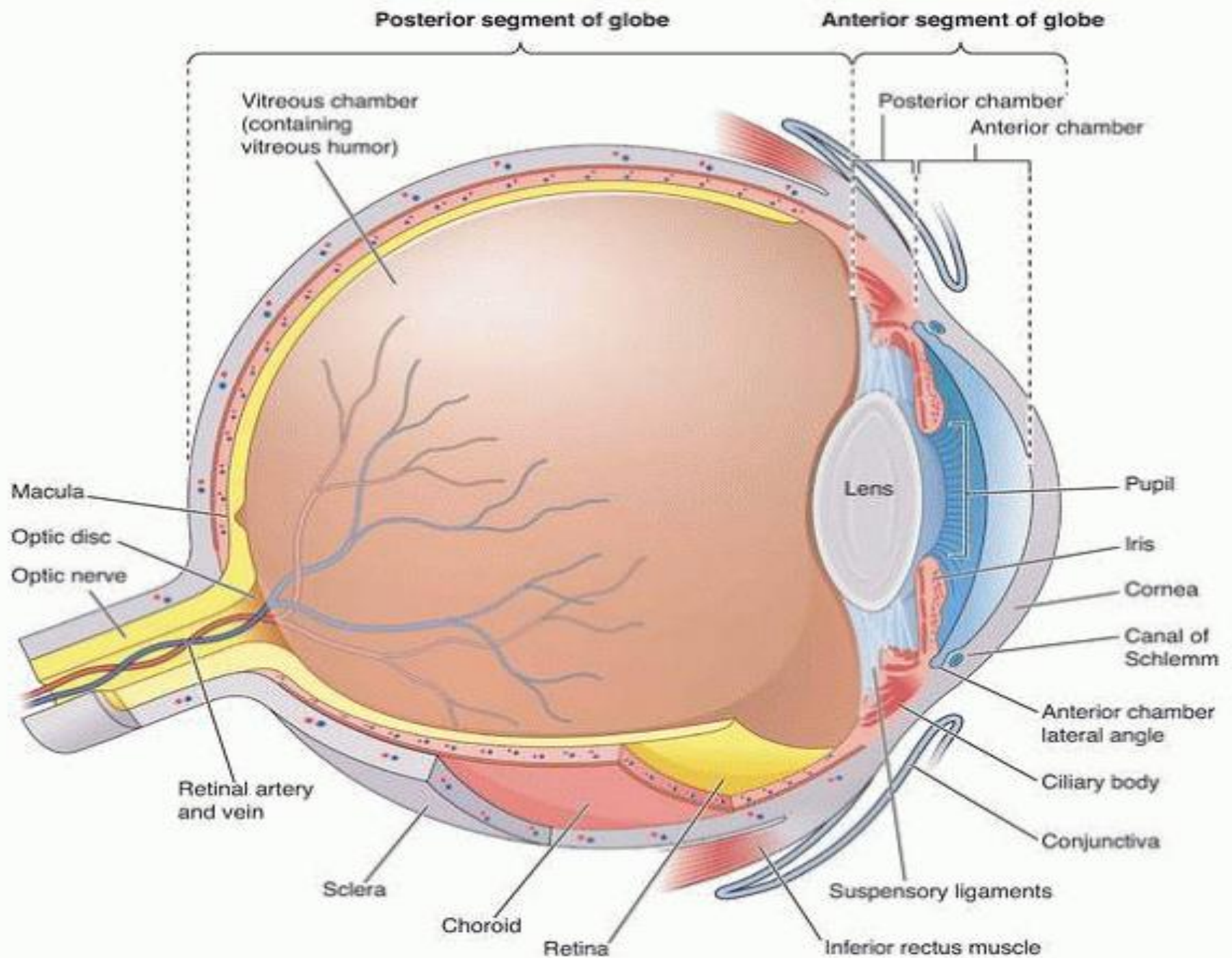
- is both sympathetic and parasympathetic
- The presynaptic parasympathetic secretomotor fibers are conveyed from the facial nerve by **the greater petrosal nerve** and then by the nerve of **the pterygoid canal** to the **pterygopalatine ganglion**,
- where they synapse with the cell body of the postsynaptic fiber.
- Vasoconstrictive, postsynaptic sympathetic fibers, brought from the **superior cervical ganglion** by the **internal carotid plexus** and **deep petrosal nerve**, join the parasympathetic fibers to form the nerve of the pterygoid canal and traverse the pterygopalatine ganglion
- **The zygomatic nerve** (from the maxillary nerve) brings both types of fibers to the lacrimal branch of the ophthalmic nerve, by which they enter the gland
- **N.B.** The postganglionic fibers leave the ganglion and join the maxillary nerve, then passes into its zygomatic branch and the zygomaticotemporal nerve.
- They reach the lacrimal gland within the lacrimal nerve.



2. Eyeball

- contains the optical apparatus of the visual system.
- It occupies most of the anterior portion of the orbit, suspended by **six extrinsic muscles** that control its movement
- It measures approximately **25 mm in diameter**.
- All anatomical structures within the eyeball have a circular or spherical arrangement.





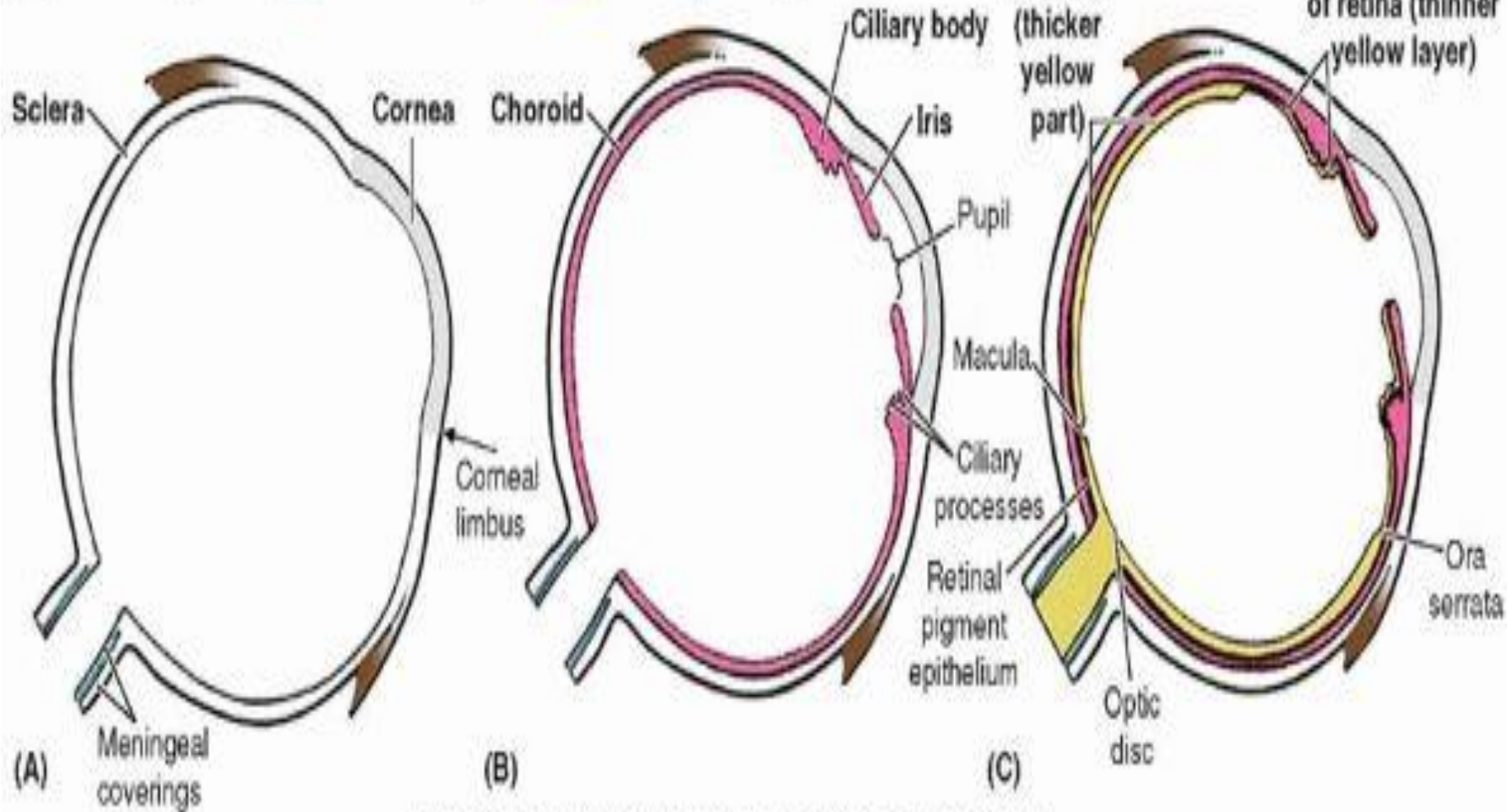
Each eye is composed of three layers:

- a tough **external layer** consisting of the **sclera** and the **cornea**
- a more **vascular middle layer** consisting of the:
 - choroid
 - ciliary body
 - and iris
- an **inner sensory layer**, the **retina**, which consists of:
 - an outer pigmented epithelium and an inner retina proper.
 - The photosensitive inner layer of the retina communicates with the cerebrum through the **optic nerve** on the eye's posterior side
 - its anterior edge is called the **ora serrata**

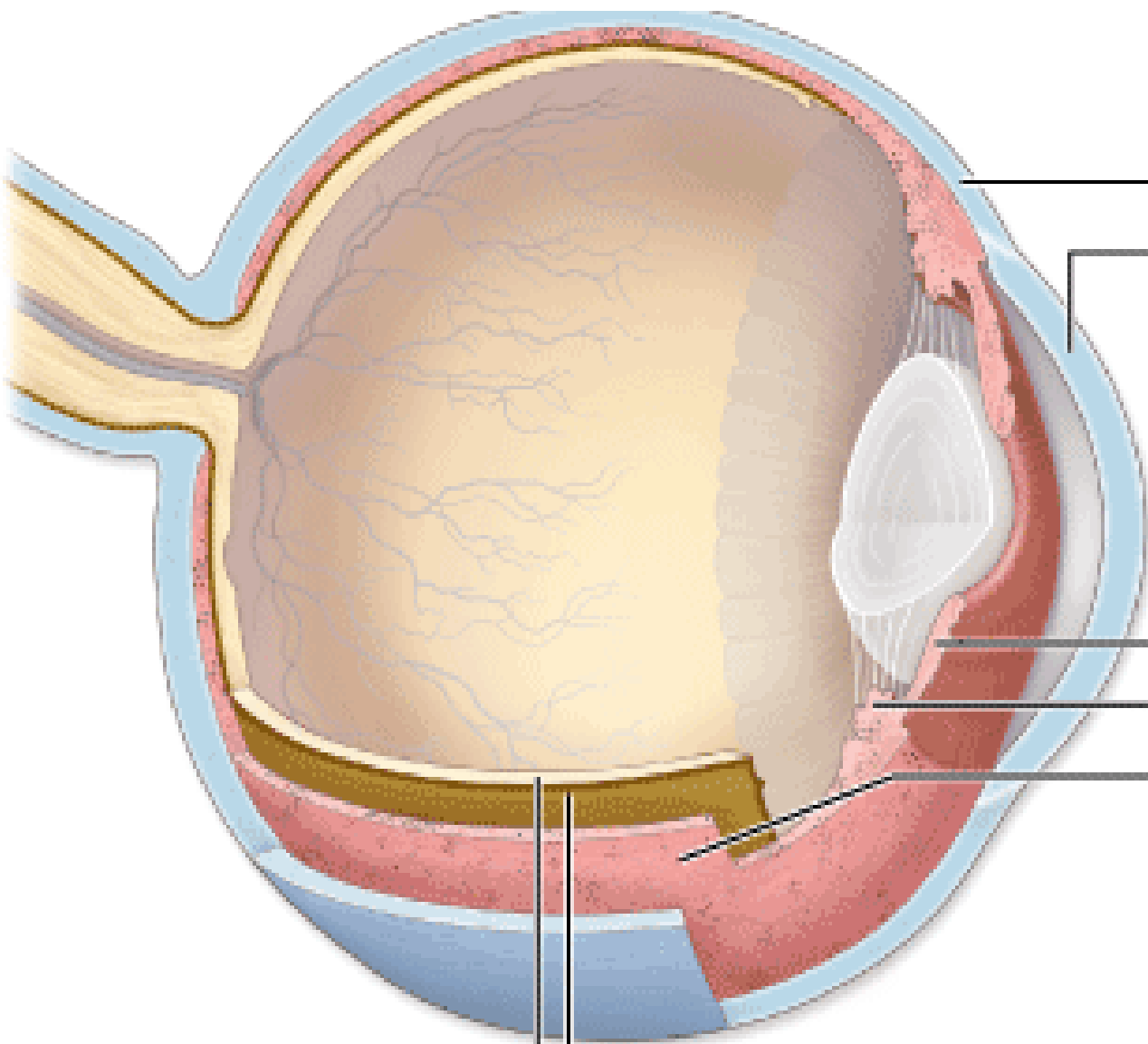
 Fibrous layer

 Vascular layer

 Inner layer of eyeball



Superior views of transversely-sectioned right eyeball



Fibrous tunic

- Sclera
- Cornea

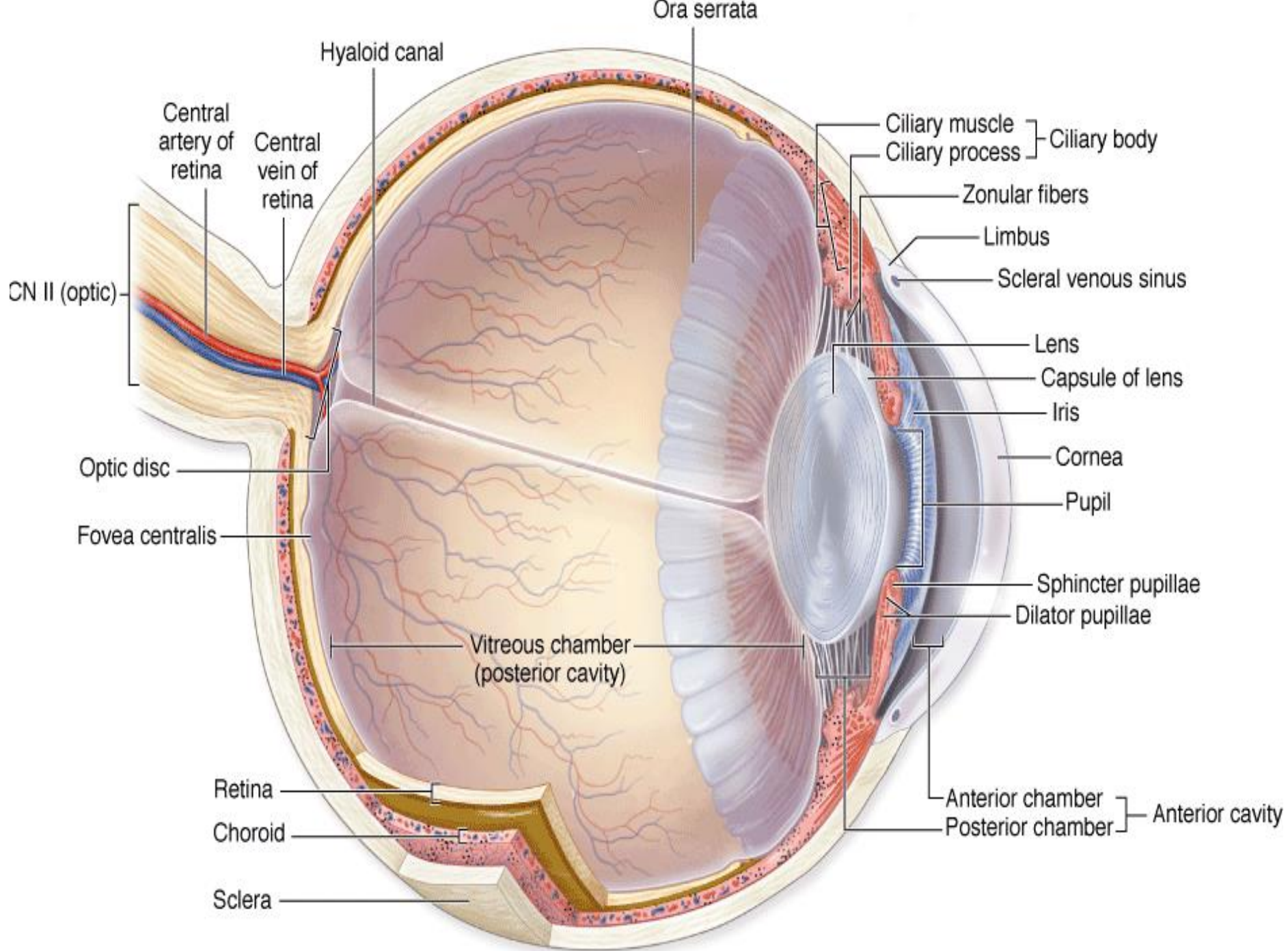
Vascular tunic

- Iris
- Ciliary body
- Choroid

Retina

- Pigmented layer
- Neural layer

- **The lens of the eye:**
 - is a biconvex transparent structure
 - held in place by a circular system of **zonular fibers**
 - which extend from the lens into a thickening of the middle layer, the **ciliary body**
 - and by close apposition to the vitreous body on its posterior side
 - Partly covering the anterior surface of the lens is an **opaque pigmented** expansion of the middle layer called the **iris**.
 - The round hole in the middle of the iris is the **pupil**



- **The eye contains two fluid-filled cavities:**
 - the anterior chamber, which occupies the space between the cornea and the iris
 - the posterior chamber, between the iris, ciliary processes, zonular attachments, and lens
 - Interconnected at the pupil, these contain a clear fluid called aqueous humor.
 - The **vitreous chamber** lies behind the lens and its zonular attachments and is surrounded by the retina.
 - This chamber is filled with a transparent, gelatinous mass of connective tissue called **the vitreous body**

MUSCLES OF THE EYE

- **Extrinsic Muscles**

- There are seven extrinsic muscles of the eye:

- the levator palpebrae superioris
- the superior, inferior, medial, and lateral recti muscles
- and the superior and inferior oblique muscles

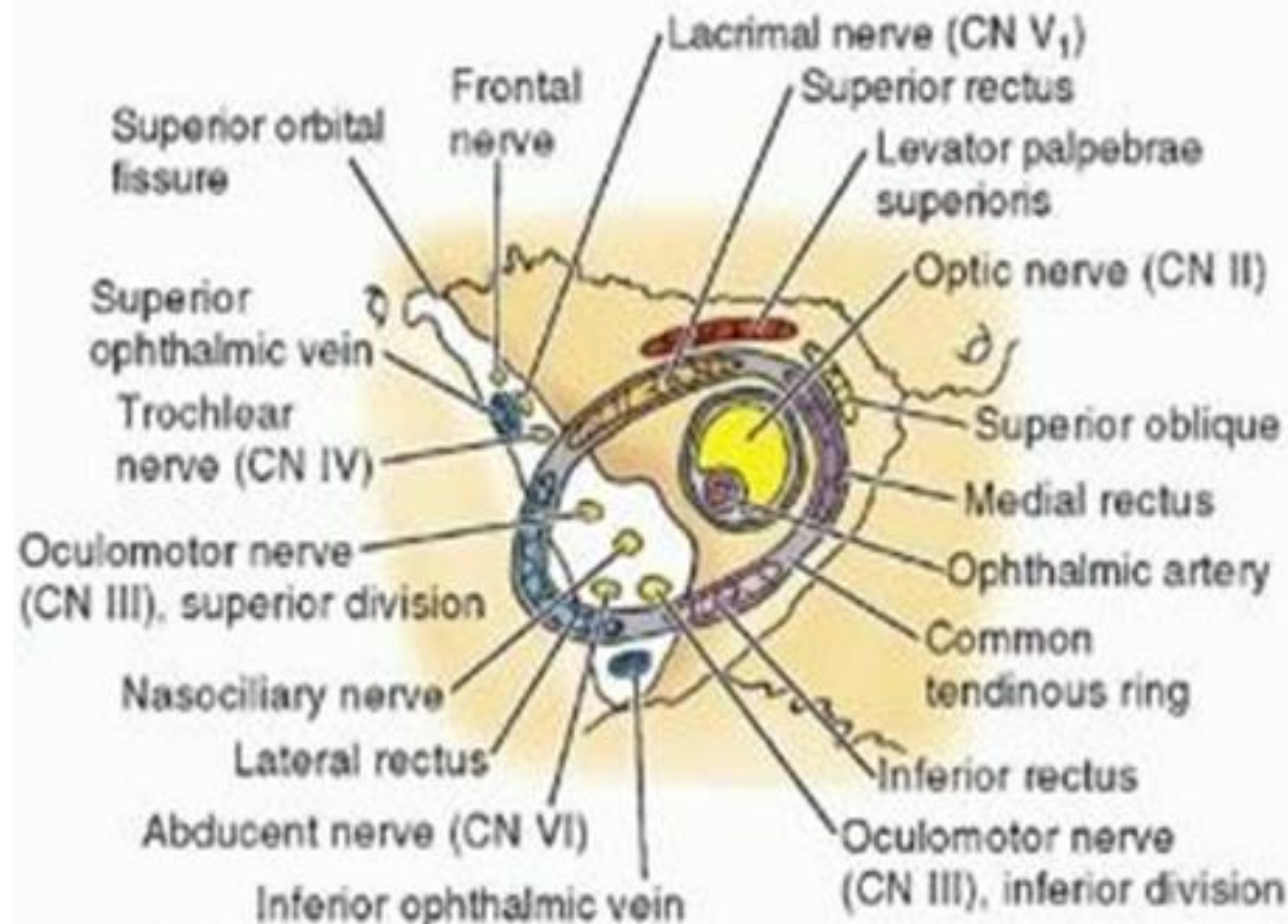
- The levator palpebrae superioris

- originates on the inferior surface of the lesser wing of the sphenoid bone
- and inserts into the upper eyelid
- this muscle functions to elevate the upper eyelid.

- The four recti muscles: lateral, medial, superior, and inferior—originate from a common tendinous ring surrounding the optic foramen

COMMON TENDINOUS RING

- is a fibrous ring which surrounds the optic canal and part of the superior orbital fissure at the apex of the orbit
- gives origin to the four recti
- The optic nerve and ophthalmic artery enter the orbit via the **optic canal, and so lie within the common tendinous ring.**
 - oculomotor nerve
 - the nasociliary branch of the ophthalmic nerve
 - the abducens nerve, enter the orbit within the common tendinous ring via the superior orbital fissure
 - The trochlear nerve and the frontal and lacrimal branches of the ophthalmic nerve all enter the orbit through the superior orbital fissure but lie outside the common tendinous ring.
 - Structures which enter the orbit through the inferior orbital fissure lie outside the common tendinous ring.



OPTIC CANAL

- OPTIC NERVE (II)
- OPHTHALMIC ARTERY

SUPERIOR ORBITAL FISSURE

ABOVE COMMON TENDINOUS RING

- TROCHLEAR NERVE (IV)
- FRONTAL AND LACRIMAL NERVES (V₁)
- SUPERIOR OPHTHALMIC VEIN
- RECURRENT MENINGEAL ARTERY

SUPERIOR ORBITAL FISSURE

WITHIN COMMON TENDINOUS RING

- OCULOMOTOR NERVES (SUP/INF) (III)
- ABDUCENS NERVE (VI)
- NASOCILIARY NERVE (V₁)

SUPERIOR ORBITAL FISSURE

BELOW COMMON TENDINOUS RING

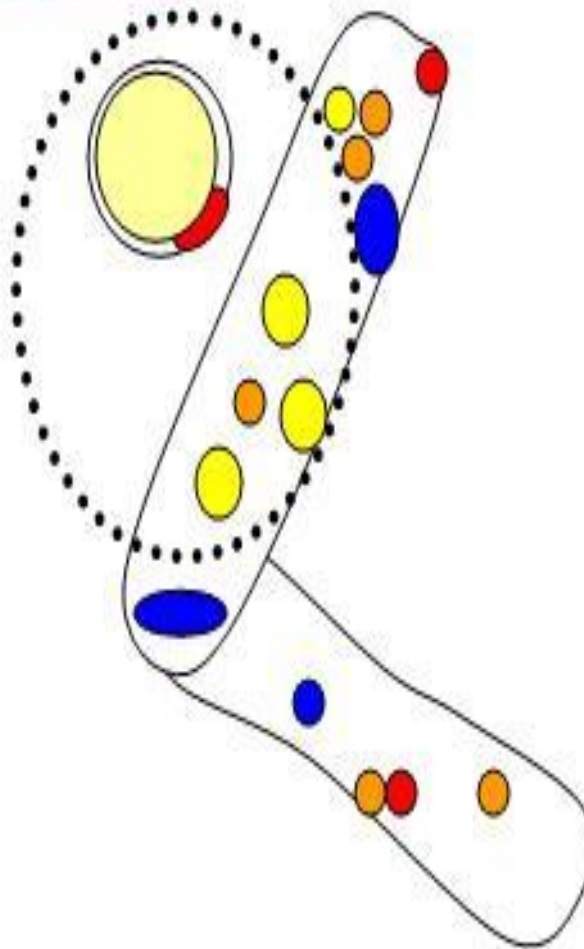
- INFERIOR OPHTHALMIC VEIN

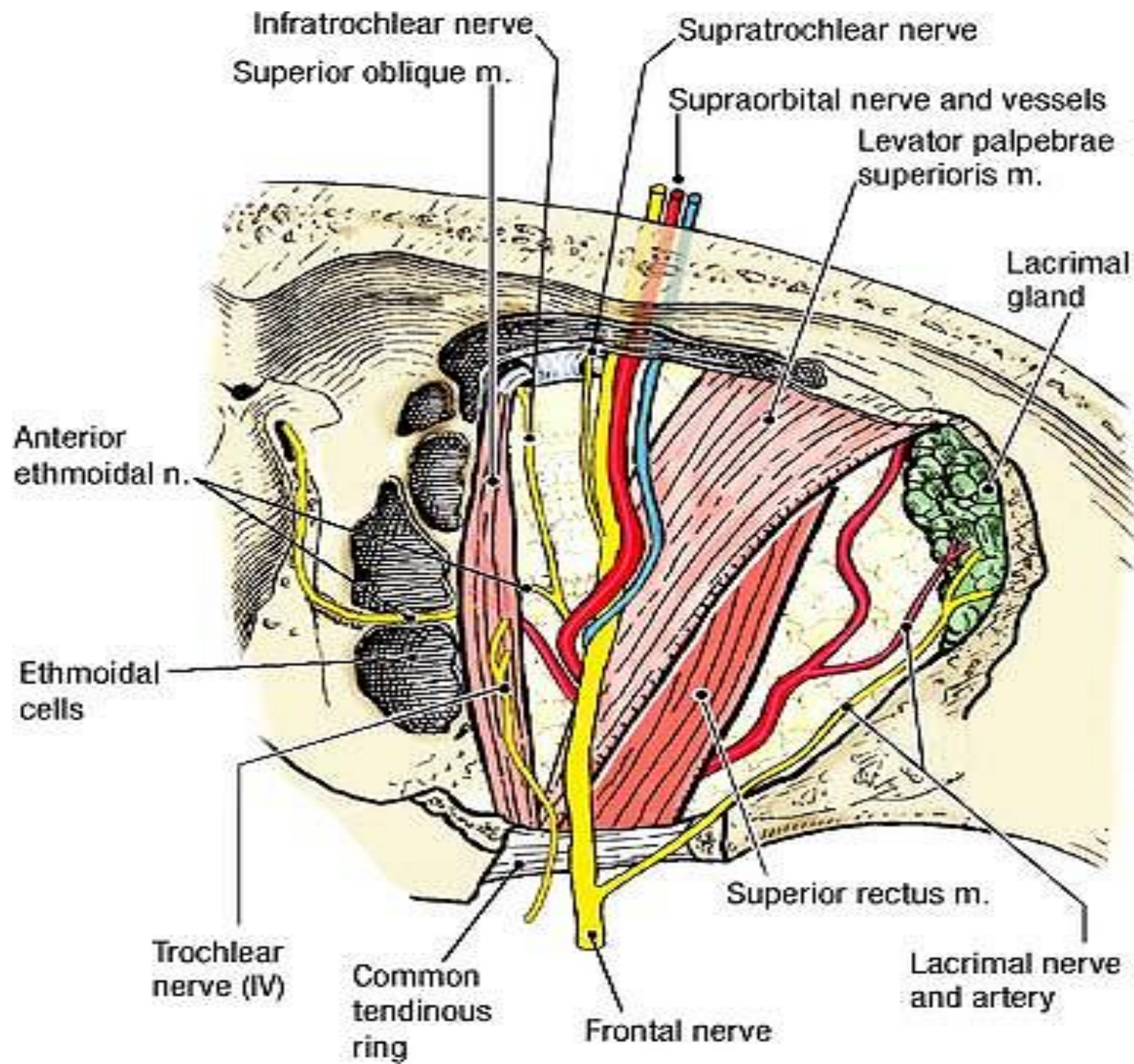
INFERIOR ORBITAL FISSURE

- INFRAORBITAL NERVE (V₂)
- INFRAORBITAL ARTERY
- ZYGOMATIC NERVE (V₂)
- VENOUS ANASTOMOSES BETWEEN
OPHTHALMIC VEIN / PTERYGOPALATINE
PLEXUS

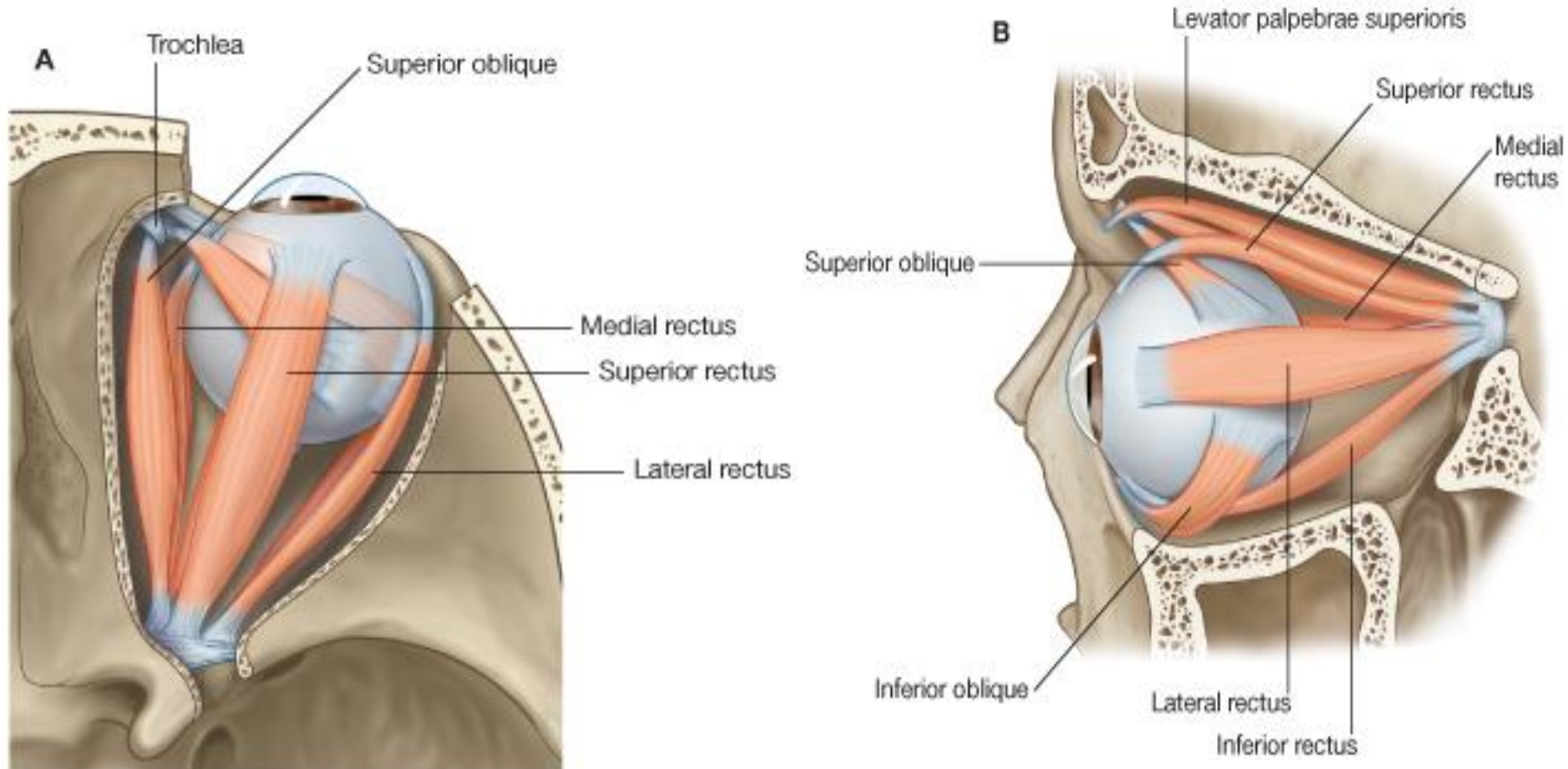
ANT / POST ETHMOIDAL FORAMEN

- ETHMOIDAL ARTERIES
- ETHMOIDAL VEINS
- ETHMOIDAL NERVES

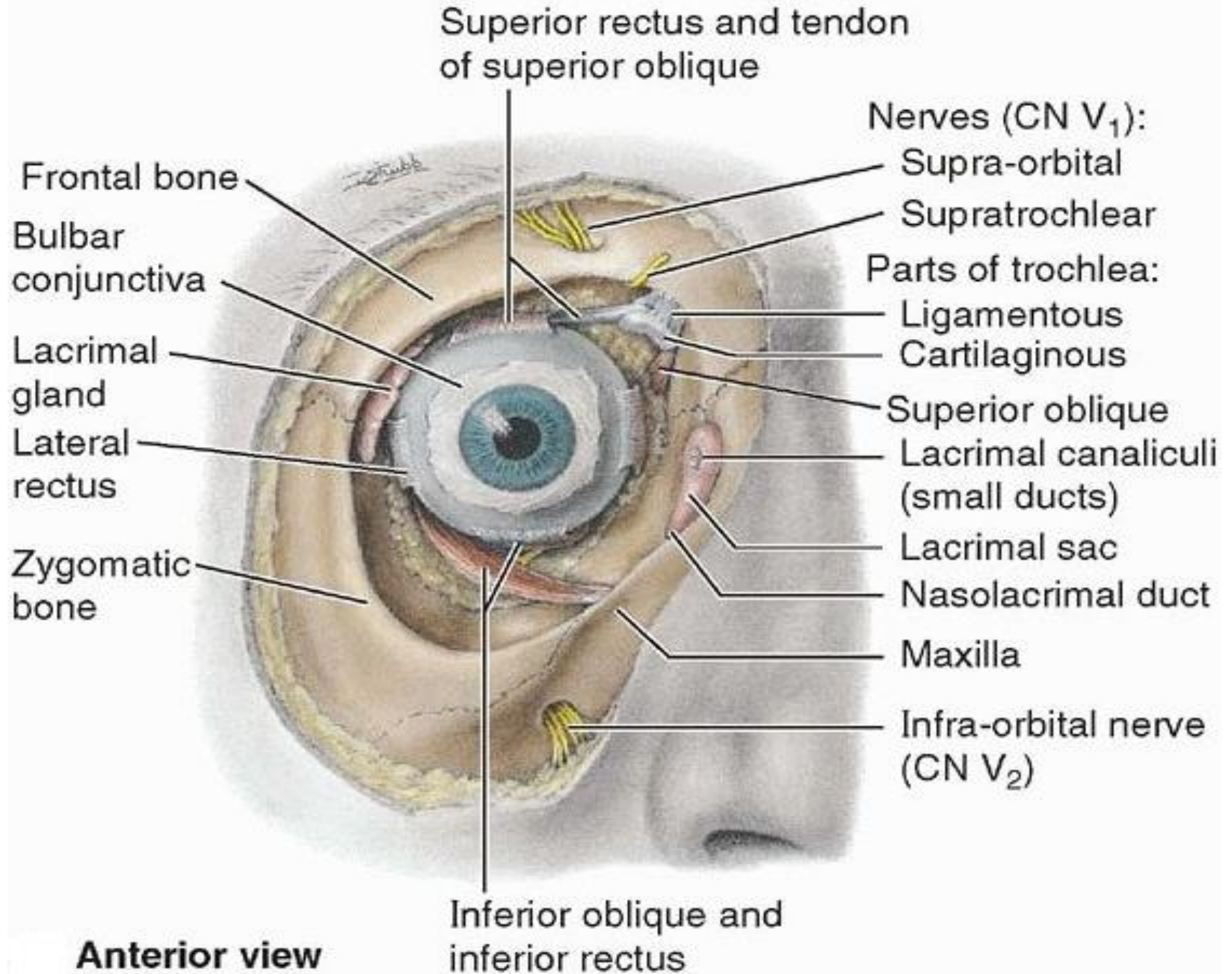


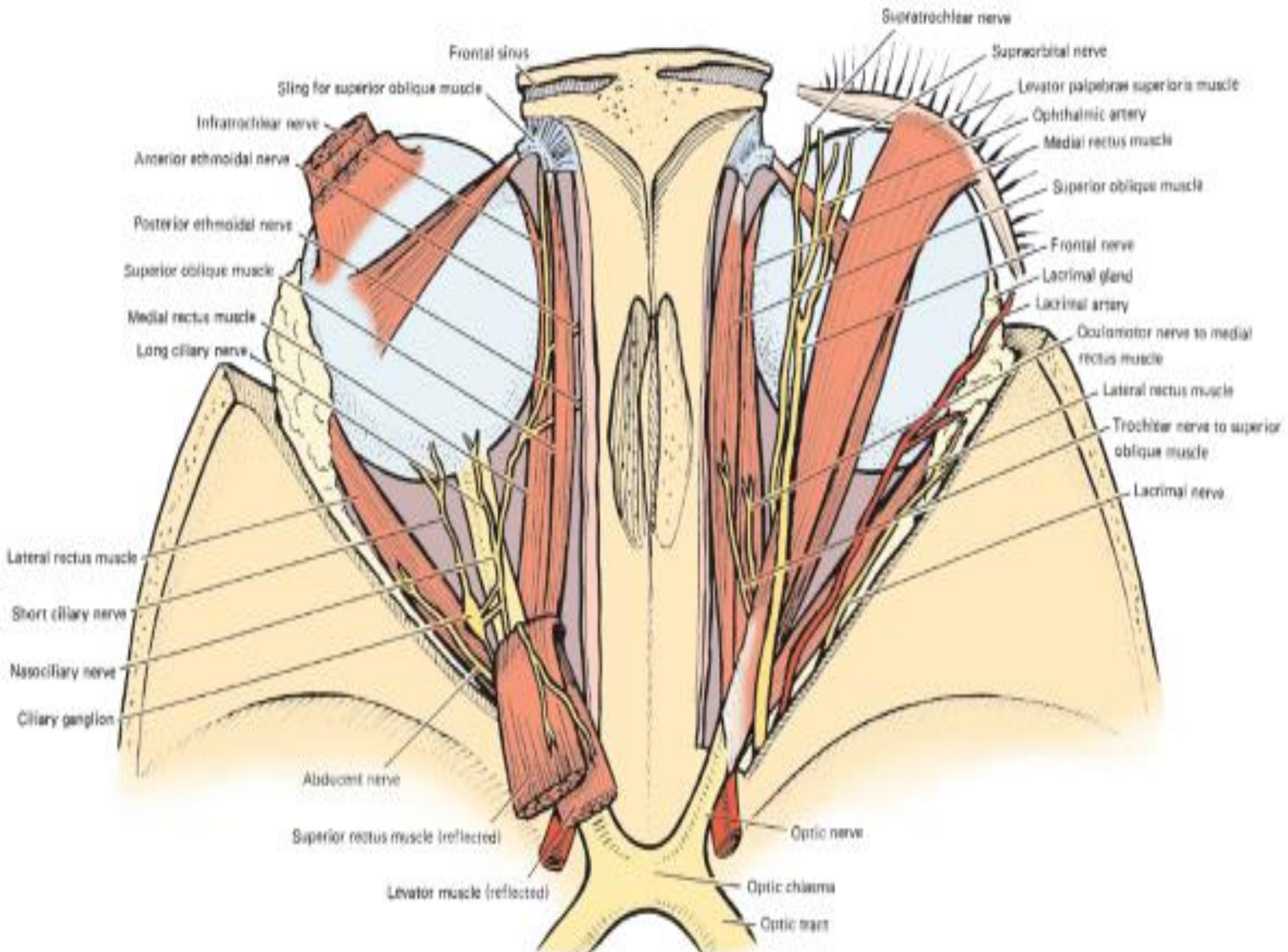


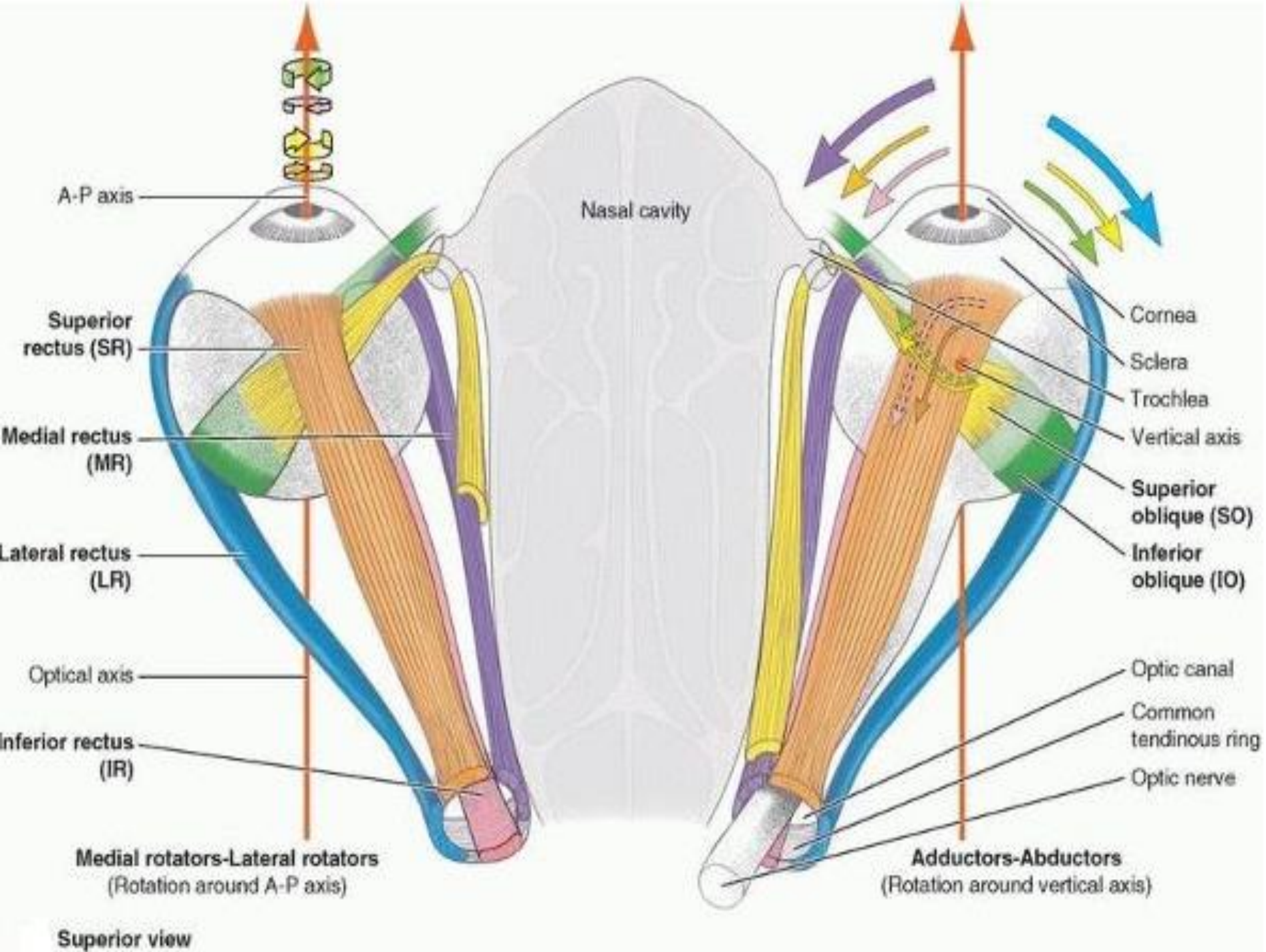
- Each of these four muscles passes into the orbit and inserts into the sclera of the orb, a few millimeters posterior to the cornea in the position indicated by their names



- **The superior oblique muscle:**
 - arises from the body of the sphenoid bone immediately above the superior rectus muscle
 - passes anteriorly to end in a tendinous pulley attached to the trochlear fovea of the frontal bone.
 - the tendon turns laterally beneath the superior rectus muscle to insert into **the sclera** between the insertions of the superior and lateral rectus muscles
 - Contraction of this muscle directs the eye downward and laterally.

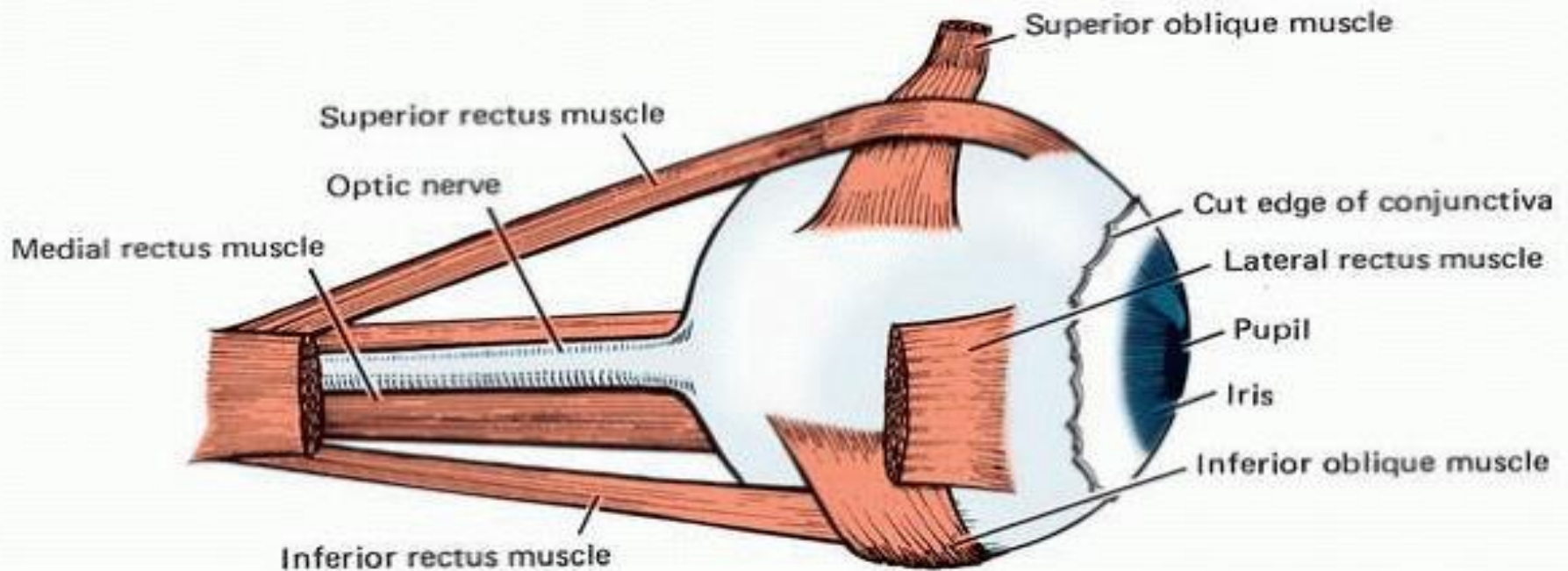






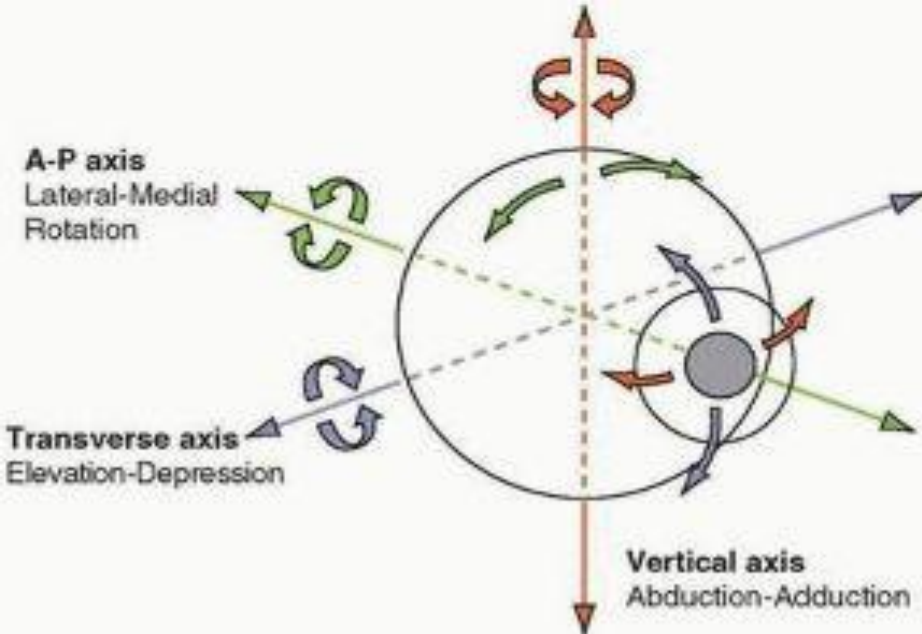
- **The inferior oblique muscle:**

- arises from the orbital surface of the **maxilla lateral to the lacrimal groove.**
- The muscle passes between the orbital floor and the inferior rectus to be inserted in the sclera between the insertions of the lateral rectus and superior oblique muscles
- The inferior oblique muscle functions to direct the eye **upward and lateralward**



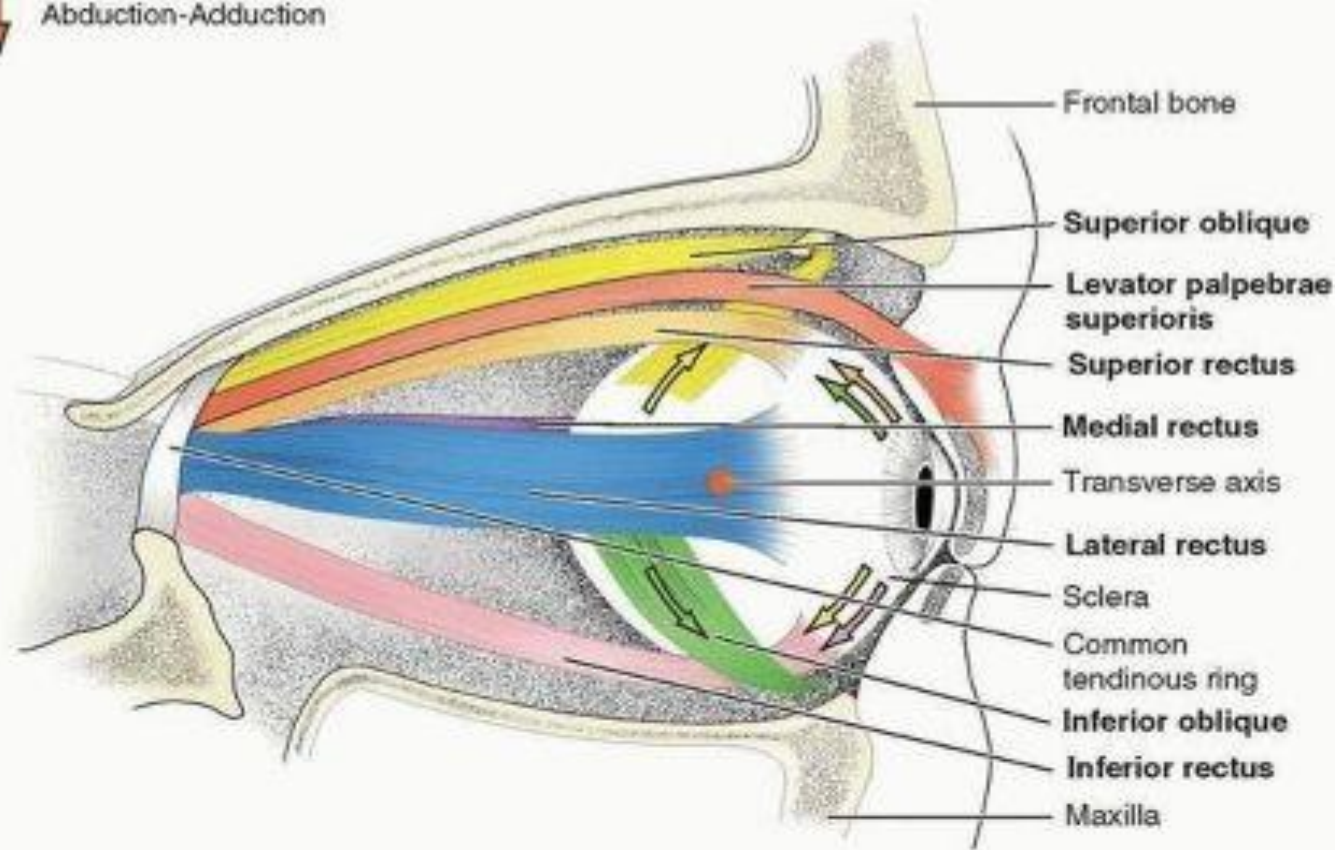
MOVEMENTS OF THE EYEBALL

- occur as rotations around three axes:
 - vertical, transverse, and anteroposterior
 - and are described according to the direction of movement of the pupil from the primary position or of the superior pole of the eyeball from the neutral position.
 - Rotation of the eyeball around the vertical axis moves the pupil medially (toward the midline, adduction) or laterally (away from the midline, abduction).
 - Rotation around the transverse axis moves the pupil superiorly (elevation) or inferiorly (depression).
 - Movements around the anteroposterior (AP) axis (corresponding to the axis of gaze in the primary position) move the superior pole of the eyeball medially (medial rotation, or intorsion) or laterally (lateral rotation, or extorsion).

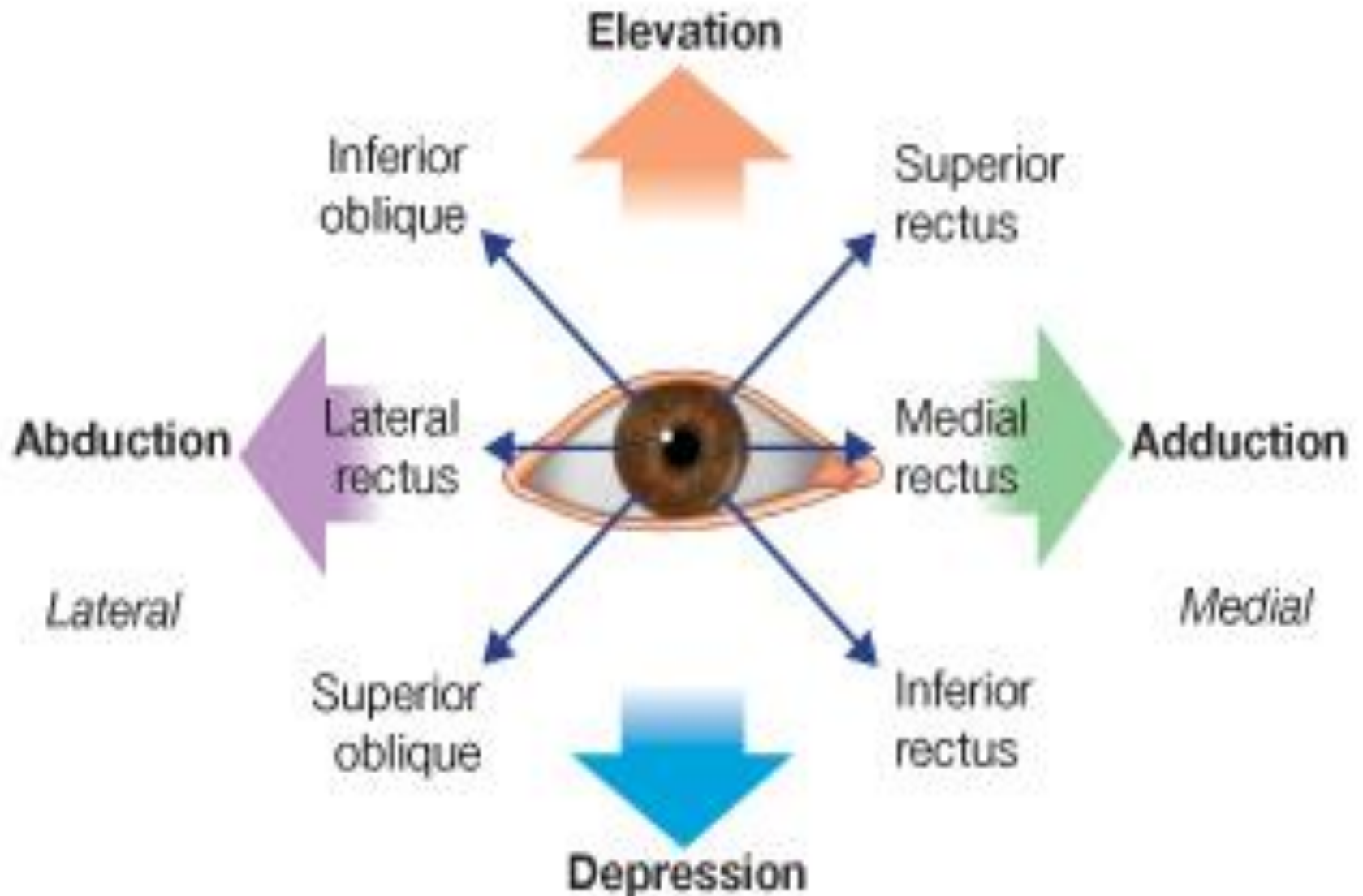


(A) Axes about which movements of the eyeball occur. Colors in (A) are NOT coordinated with B-D or Table 7.8.

(B) Lateral view
Elevators-Depressors
(Rotation around transverse axis)



Movement of the eye



Muscle tested

**Direction to move
eye when testing
muscle**

Superior rectus



Look laterally
and **upward**

Inferior rectus



Look laterally
and **downward**

Lateral rectus



Look laterally

Medial rectus



Look **medially**

Inferior oblique



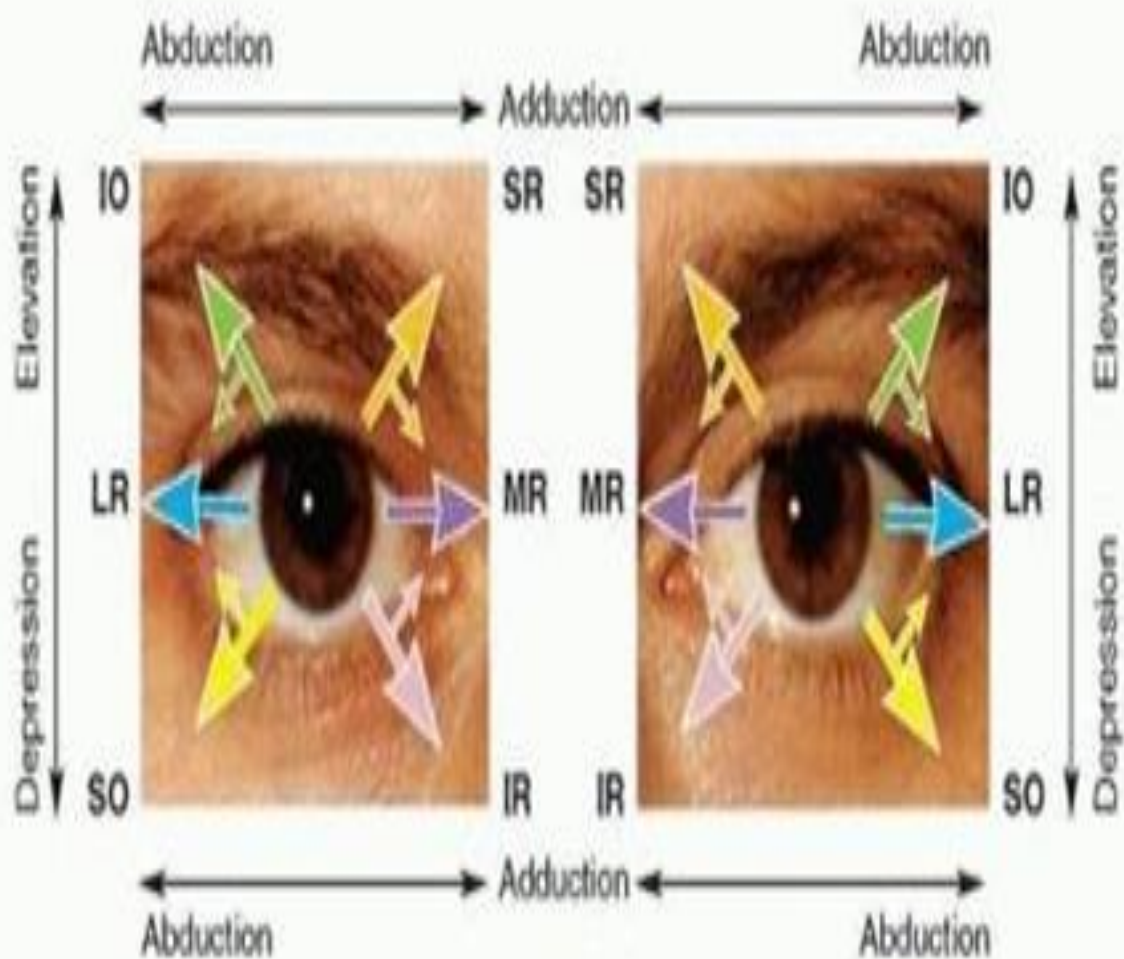
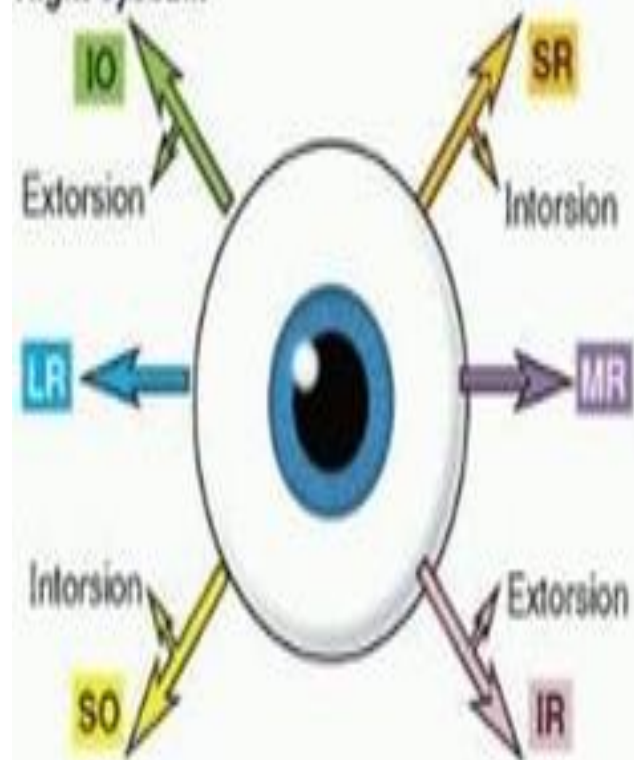
Look medially
and **upward**

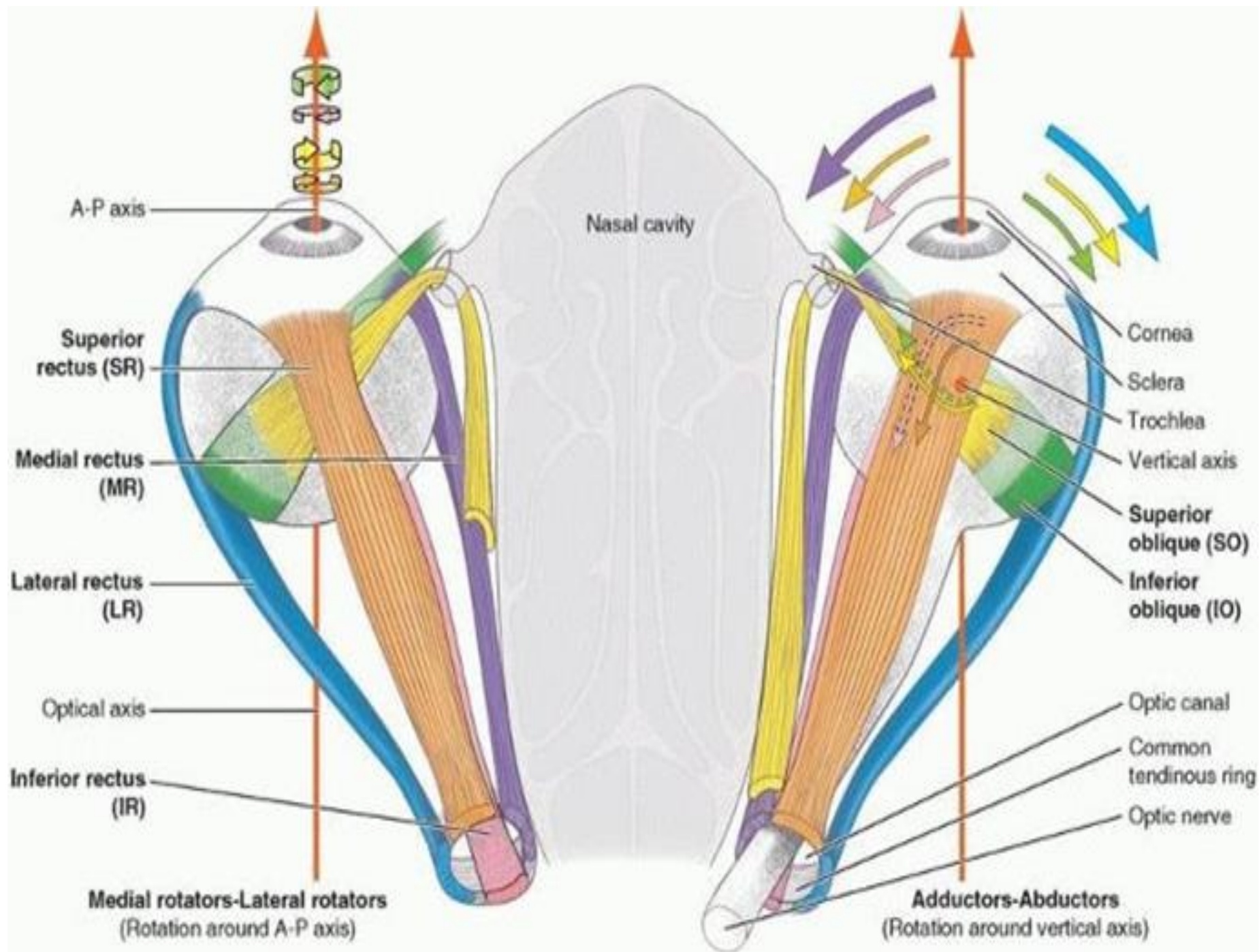
Superior oblique



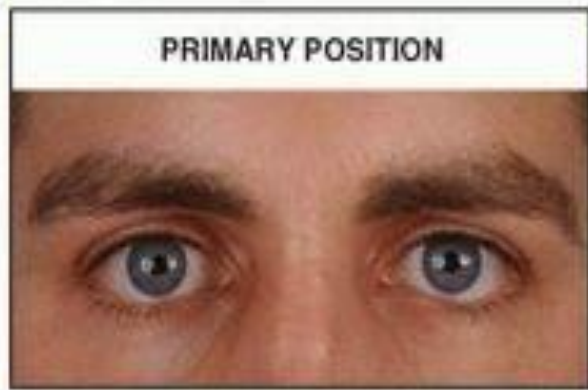
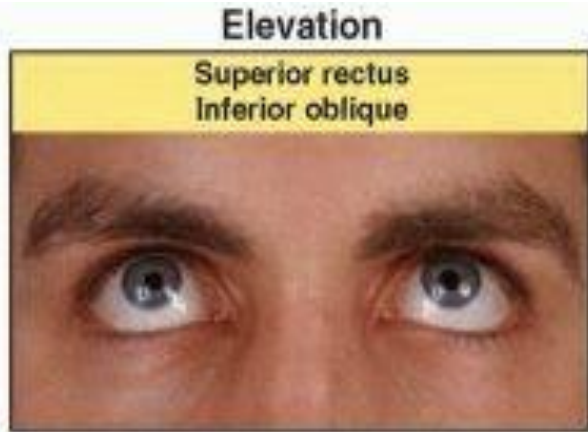
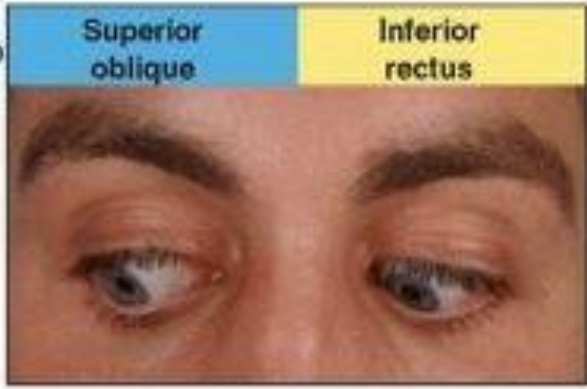
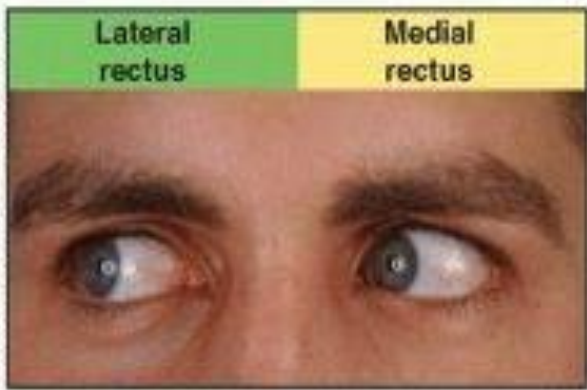
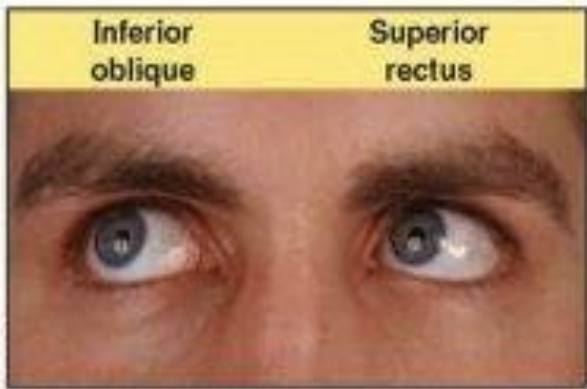
Look medially
and **downward**

Right eyeball:

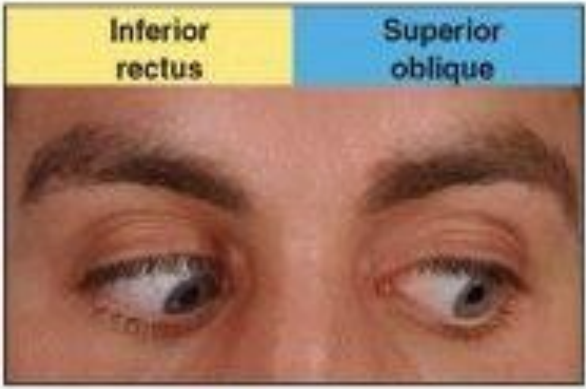
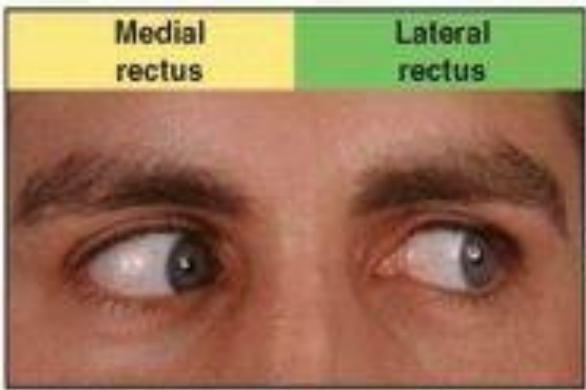
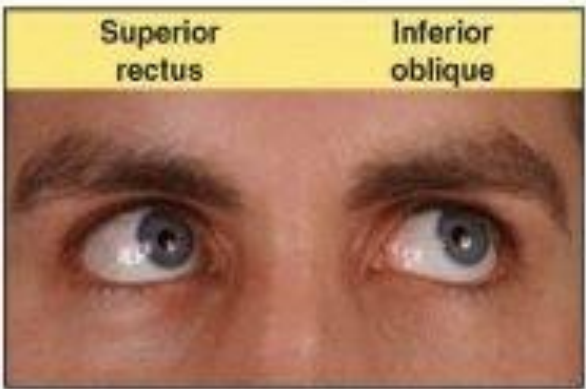




Right Abduction - Left Adduction



Depression



Left Abduction - Right Adduction

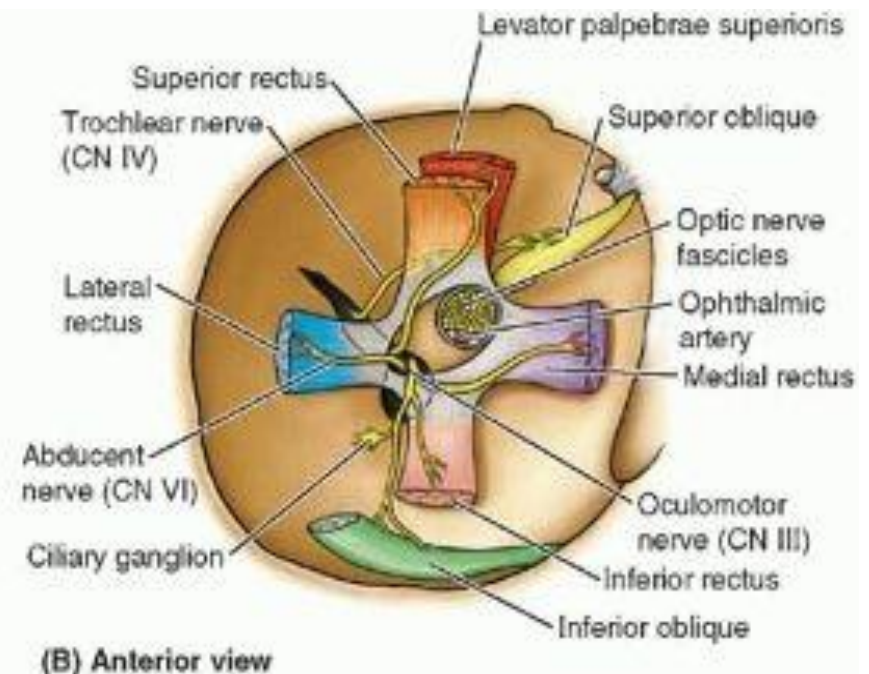
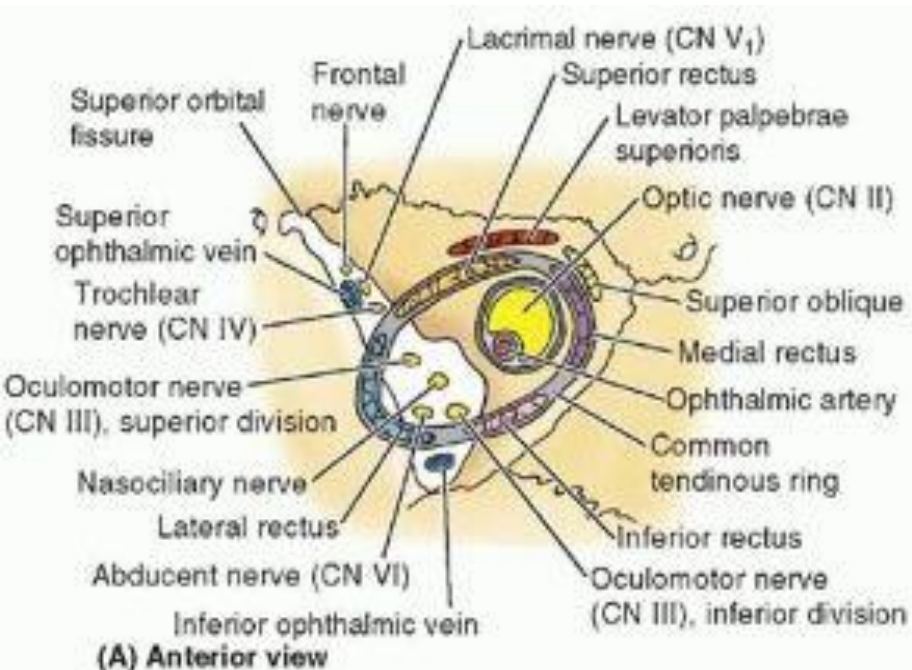
- Oculomotor nerve (CN III)
- Trochlear nerve (CN IV)
- Abducent nerve (CN VI)

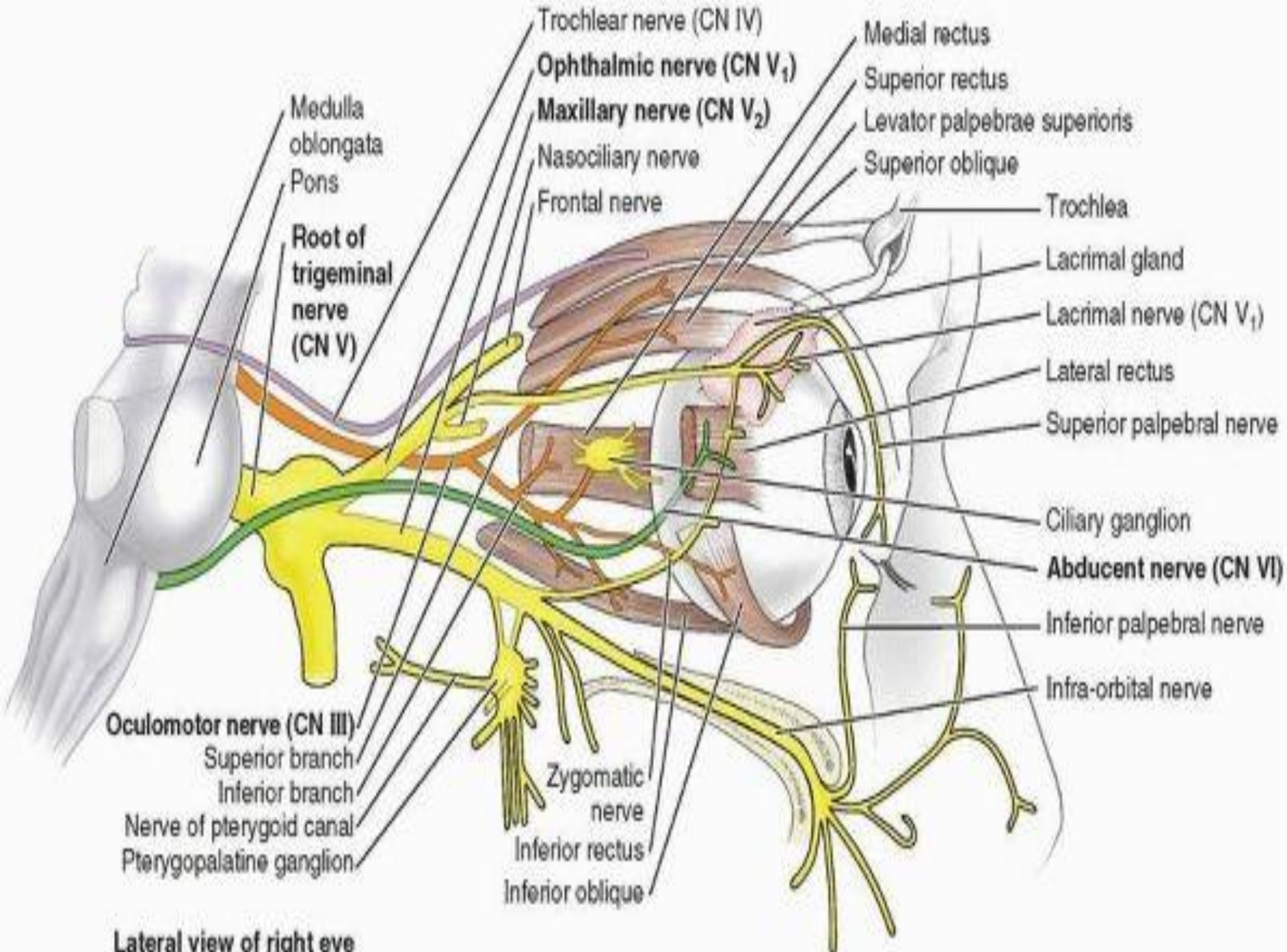
Innervation:

- The levator palpebrae superioris, the superior, inferior, and medial recti, and the inferior oblique muscles are **all innervated by branches of the oculomotor nerve.**
- The superior oblique muscle is innervated by the trochlear nerve, the sole function of this nerve.
- Innervation to the lateral rectus muscle is provided by the abducent nerve, another cranial nerve that serves only a single function

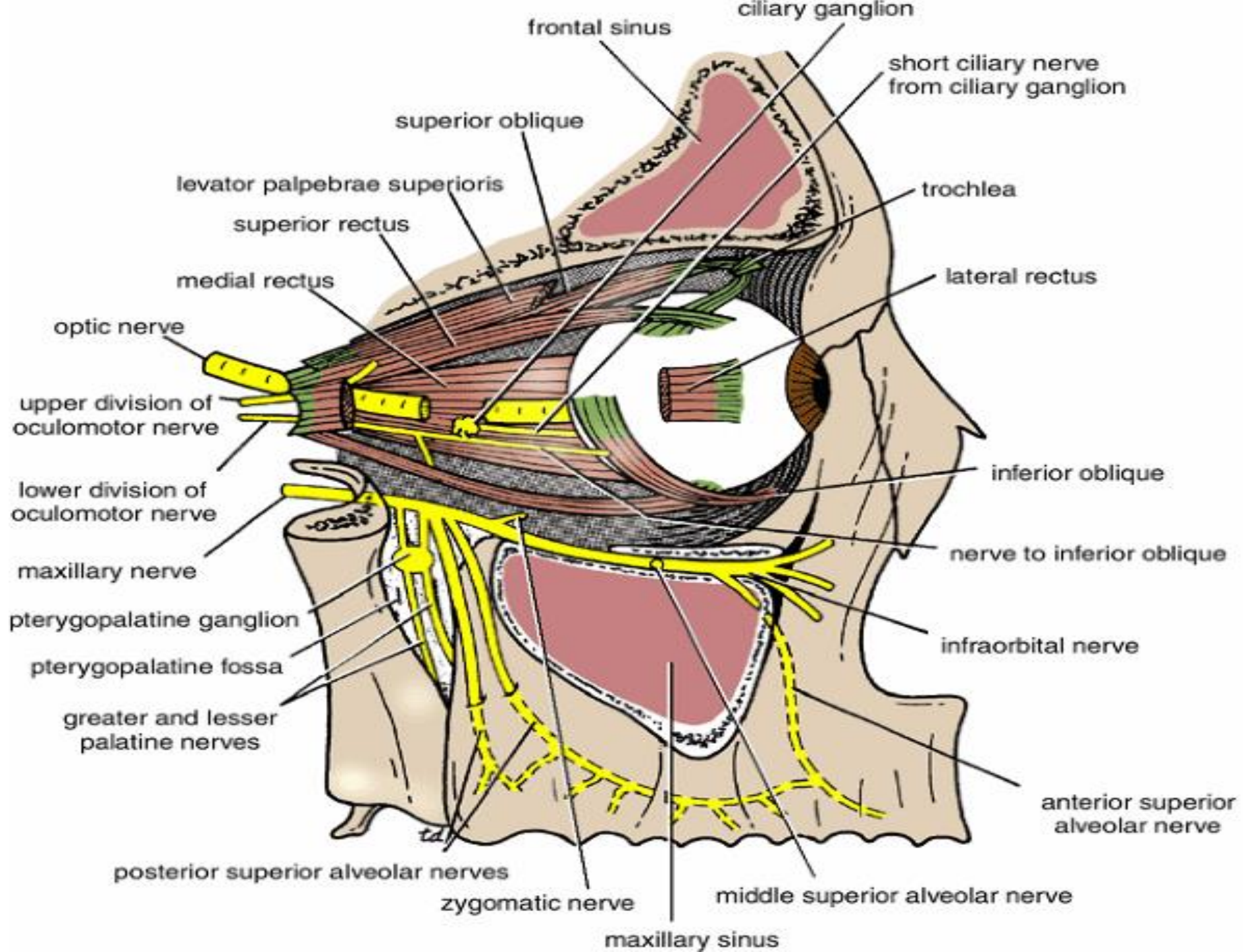
Nerves of the Orbit

- The orbit possesses nerve components from six cranial nerves:
 - the optic,
 - oculomotor, trochlear, trigeminal, abducent
 - and secretomotor fibers from the facial.
 - In addition, sympathetic fibers also serve the orbit



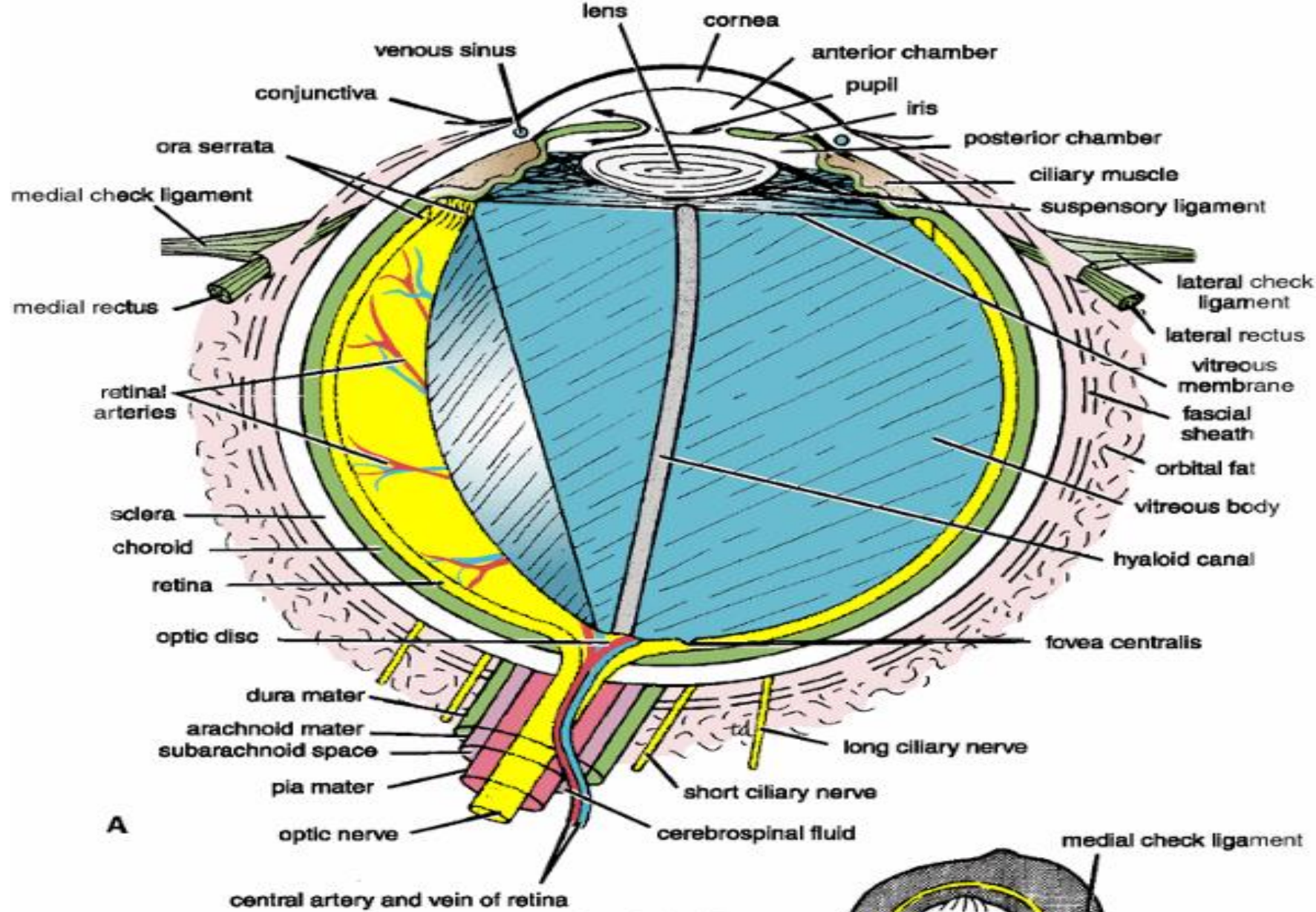


Lateral view of right eye

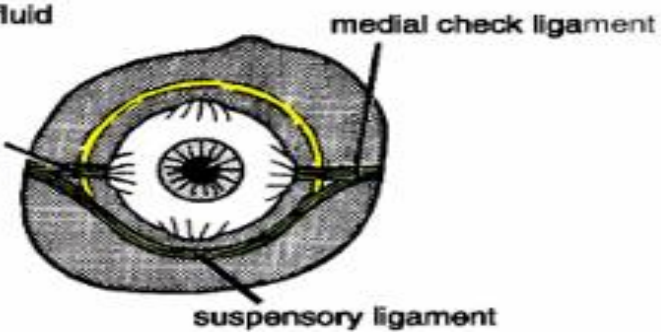


Optic Nerve

- enters the orbit from the middle cranial fossa by passing through the optic canal
- is accompanied by the ophthalmic artery, which lies on its lower lateral side.
- The nerve is surrounded by sheaths of pia mater, arachnoid mater, and dura mater
- It runs forward and laterally within the cone of the recti muscles and pierces the sclera at a point medial to the posterior pole of the eyeball.
- Here, the meninges fuse with the sclera so that the subarachnoid space with its contained cerebrospinal fluid extends forward from the middle cranial fossa, around the optic nerve, and through the optic canal, as far as the eyeball.
- A rise in pressure of the cerebrospinal fluid within the cranial cavity therefore is transmitted to the back of the eyeball



B

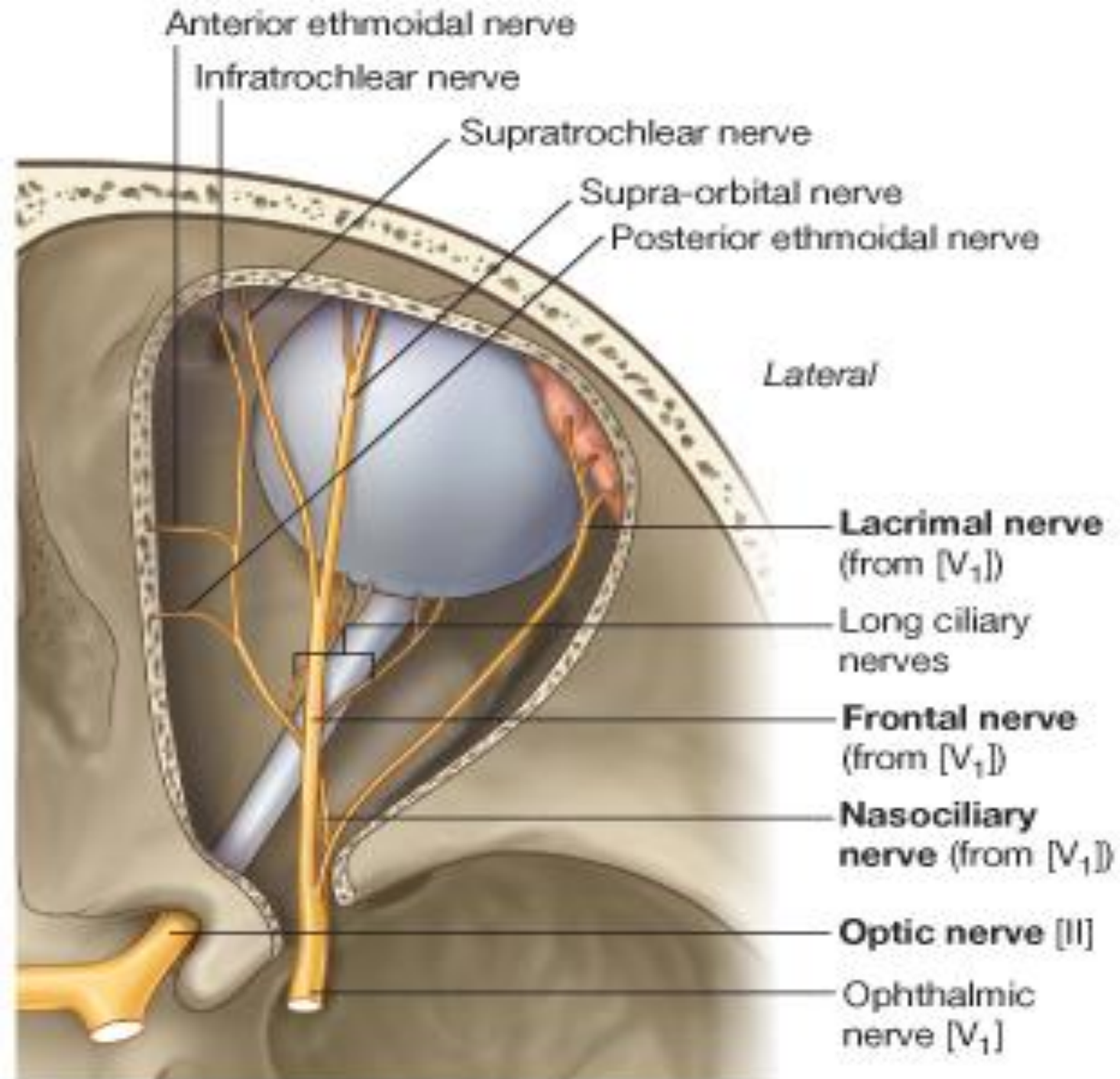


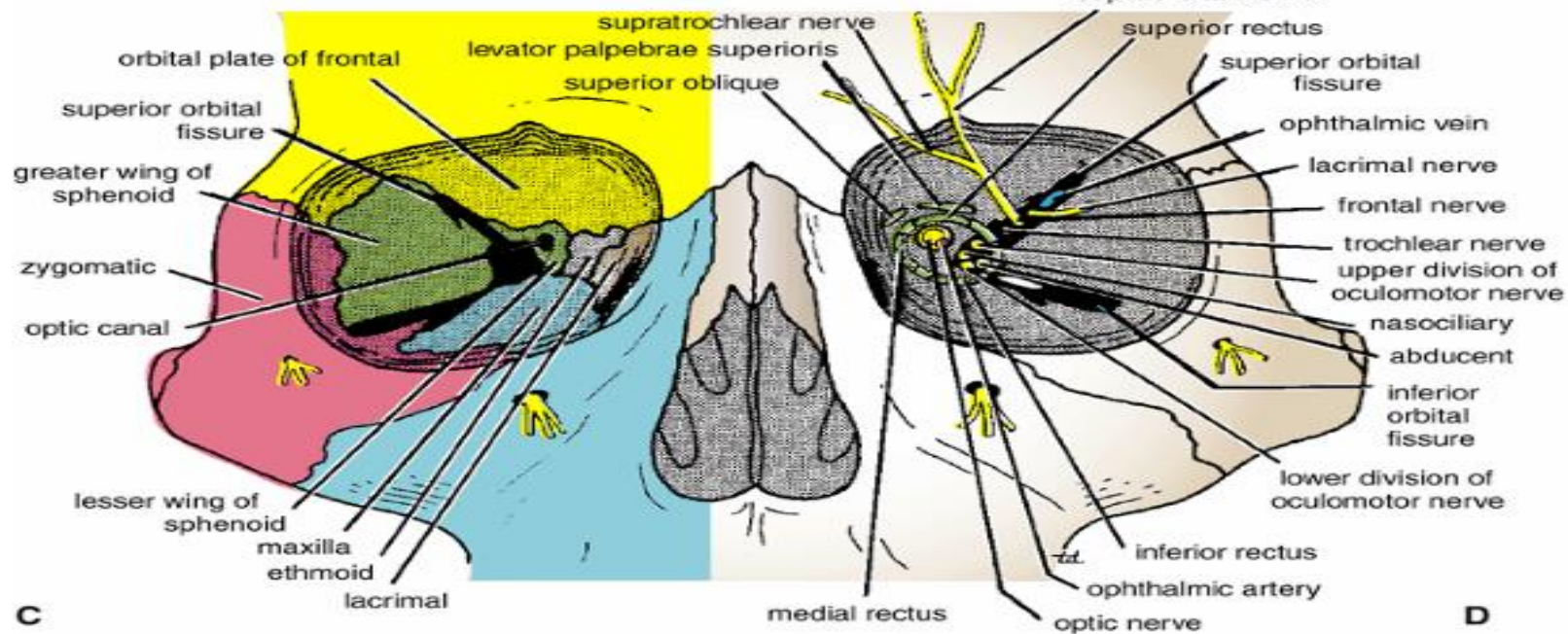
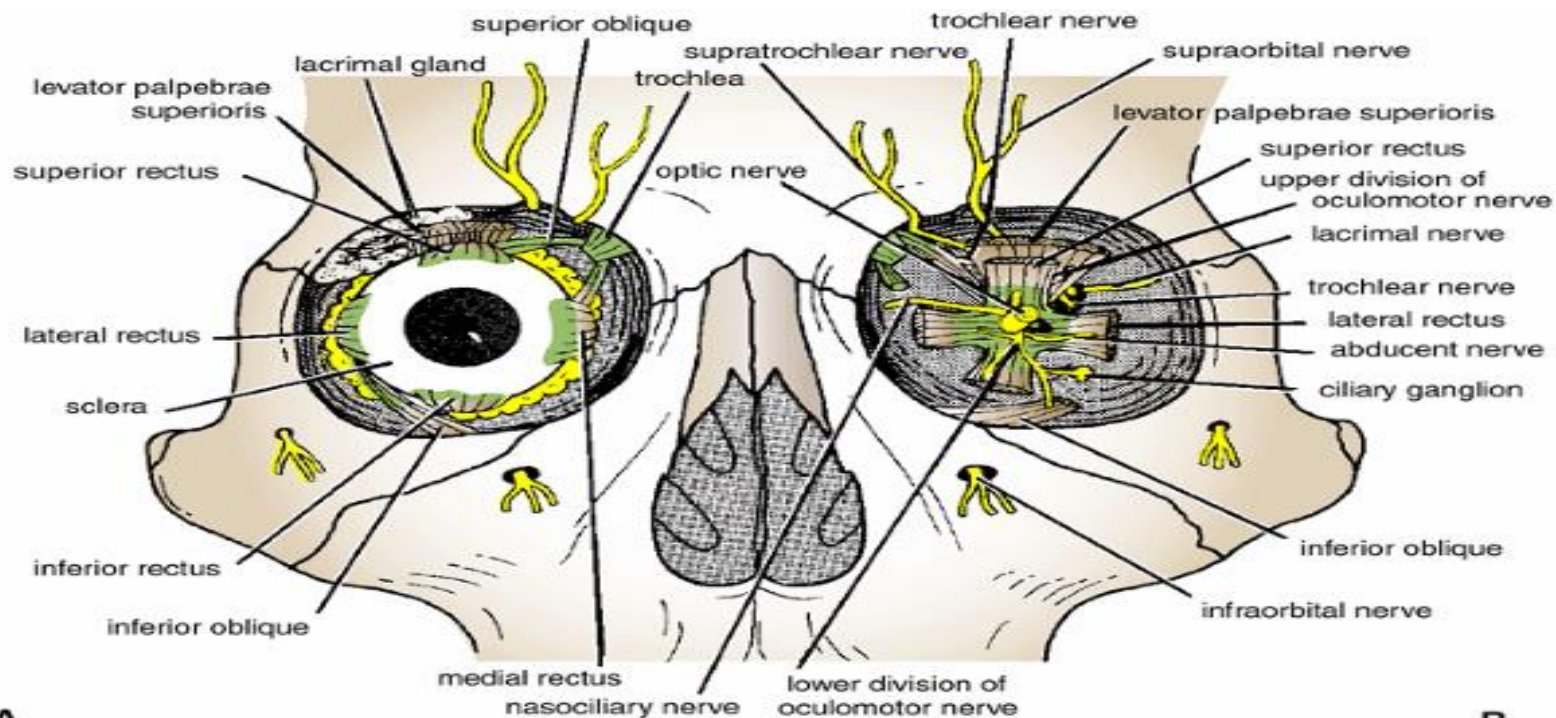
Lacrimal Nerve

- arises from the ophthalmic division of the trigeminal nerve.
- It enters the orbit through the upper part of the superior orbital fissure
- and passes forward along the upper border of the lateral rectus muscle
- It is joined by a branch of the zygomaticotemporal nerve, which later leaves it to enter the lacrimal gland (parasympathetic secretomotor fibers).
- The lacrimal nerve ends by supplying the skin of the lateral part of the upper l

Frontal Nerve

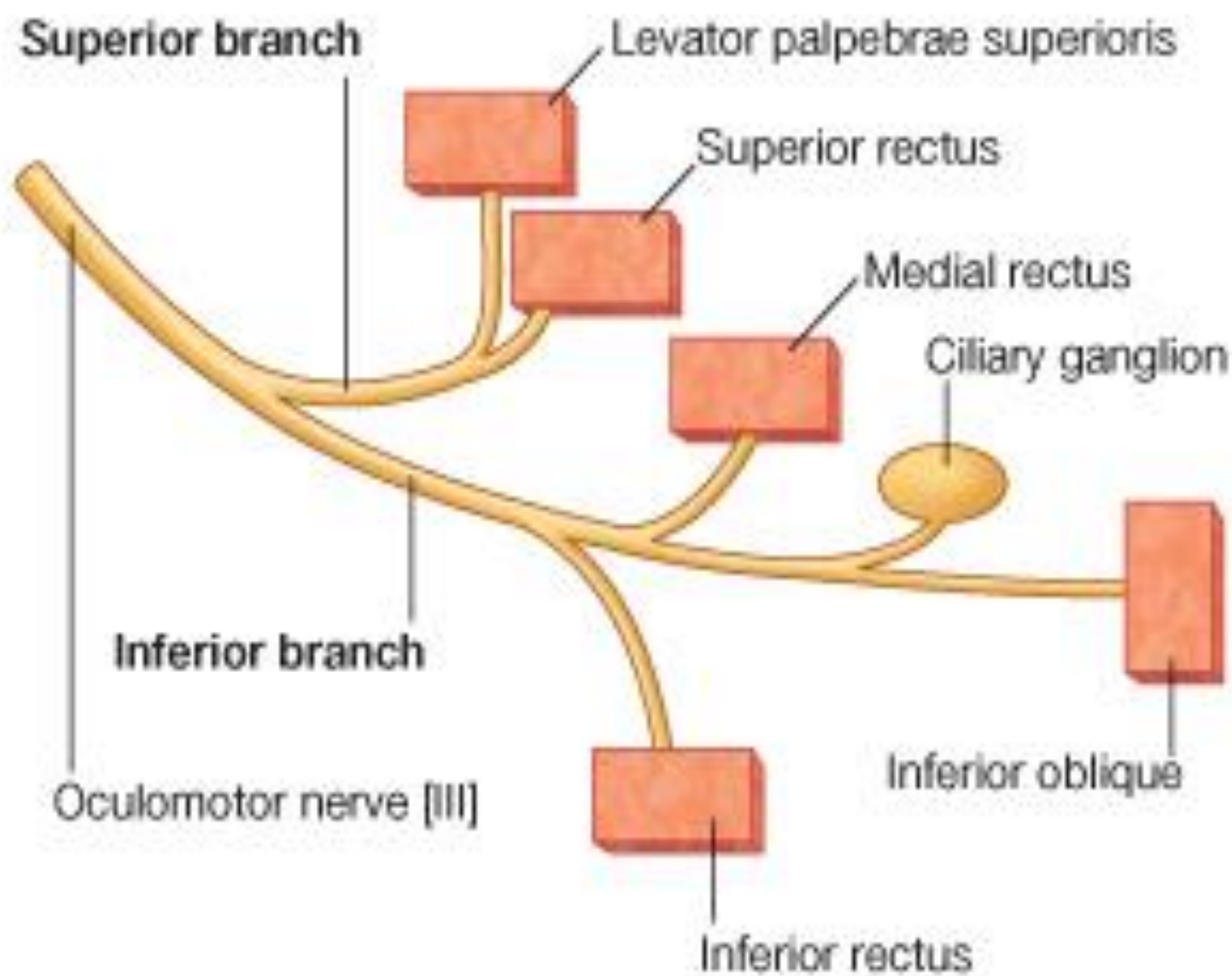
- arises from the ophthalmic division of the trigeminal nerve.
- It enters the orbit through the upper part of the superior orbital fissure
- and passes forward on the upper surface of the levator palpebrae superioris beneath the roof of the orbit
- It divides into the supratrochlear and supraorbital nerves that wind around the upper margin of the orbital cavity to supply the skin of the





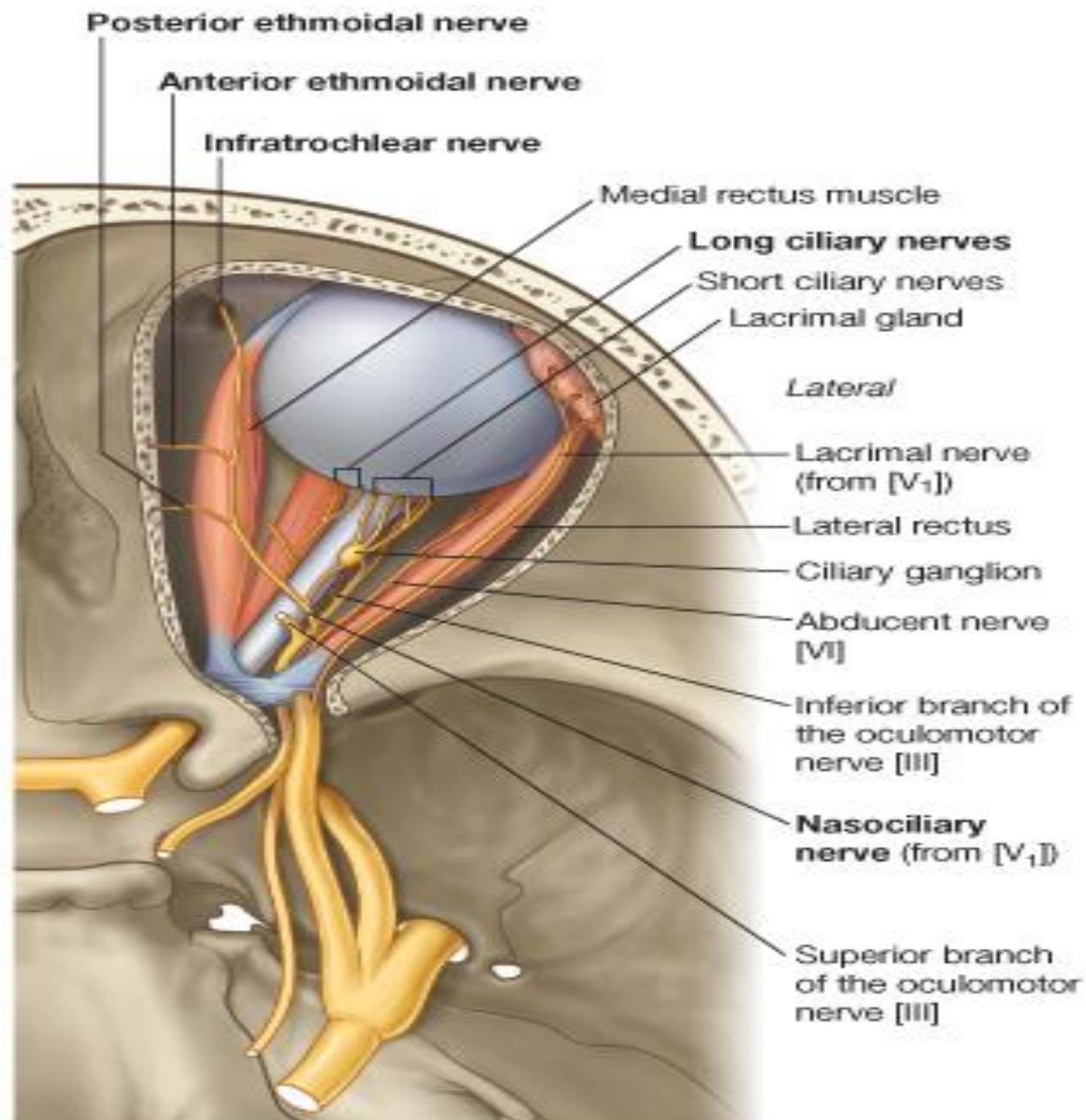
Oculomotor Nerve

- enters the orbit through the superior orbital fissure
- it divides into **superior and inferior** divisions to innervate the intrinsic ciliary and sphincter pupillae muscles and all of the extrinsic muscles except **the lateral rectus** and **the superior oblique**.
- The superior division sends branches to the levator palpebrae superioris and the superior rectus
- whereas the inferior division innervates the medial and inferior recti and the inferior oblique muscles.

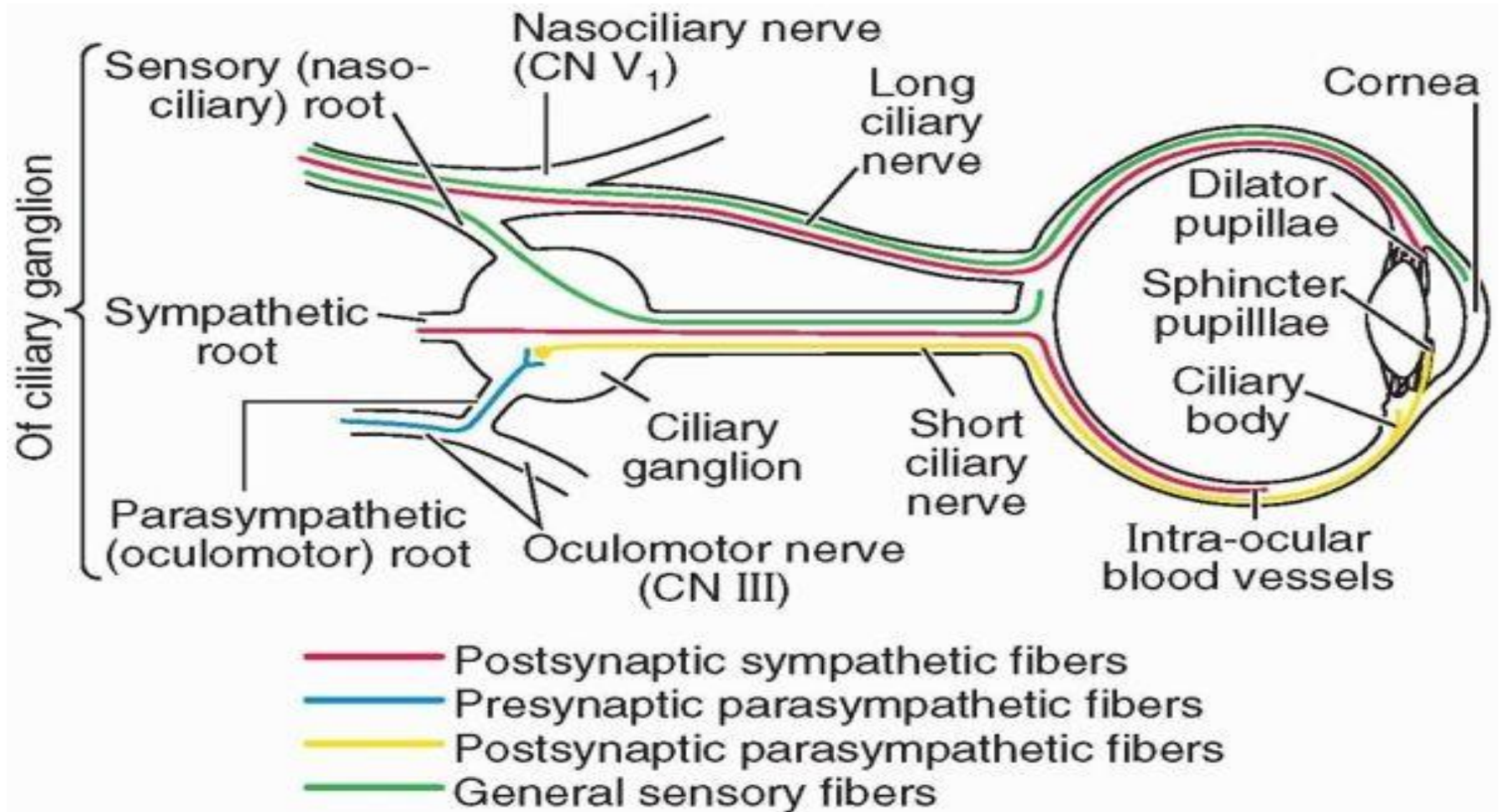


- **The inferior division** supplies the medial and inferior rectus and inferior oblique and carries **presynaptic parasympathetic fibers** to the ciliary ganglion - a parasympathetic terminal ganglion located about **1 cm** from the apex of the orbit between the optic nerve and the lateral rectus muscle.
 - Preganglionic parasympathetic fibers synapse on postganglionic cell bodies within the ganglion.
 - The axons of these cell bodies reach the orb via the **short ciliary nerves**, to be distributed to the ciliary and sphincter pupillae muscles

- **The ciliary ganglion:**
 - is a small group of postsynaptic parasympathetic nerve cell bodies associated with CN V₁
 - It is located between the optic nerve and the lateral rectus toward the posterior limit of the orbit.
 - **The ganglion receives nerve fibers from three sources:**
 - Sensory fibers from CN V₁ via the sensory or nasociliary root of the ciliary ganglion).
 - Presynaptic parasympathetic fibers from CN III via the parasympathetic or oculomotor root of the ciliary ganglion.
 - Postsynaptic sympathetic fibers from the internal carotid plexus via the sympathetic root of the ciliary ganglion



- The short ciliary nerves:
 - arise from the ciliary ganglion and are considered to be branches of CN V₁
 - They carry parasympathetic and sympathetic fibers to the ciliary body and iris.
 - The short ciliary nerves consist of postsynaptic parasympathetic fibers originating in the ciliary ganglion, afferent fibers from the nasociliary nerve that pass through the ganglion
 - and postsynaptic sympathetic fibers that also pass through it.
- Long ciliary nerves:
 - branches of the nasociliary nerve (CN V₁) that pass to the eyeball, bypassing the ciliary ganglion, convey postsynaptic sympathetic fibers to the dilator pupillae and afferent fibers from the iris and cornea.



N.B. All parasympathetic innervation but only some of the sensory and sympathetic innervation to the eyeball traverses the ganglion. Sympathetic and sensory fibers in the long ciliary nerve bypass the ganglion.

Vasculature of Orbit

- ARTERIES OF ORBIT
- The blood supply of the orbit is mainly from the **ophthalmic artery**

In the orbit the ophthalmic artery gives off:

- the lacrimal artery:
 - arises from the ophthalmic artery on the lateral side of the optic nerve
 - passes anteriorly on the lateral side of the orbit, supplying the lacrimal gland, muscles, the anterior ciliary branch to the eyeball, and the lateral sides of the eyelid
- The central retinal artery:
 - arising inferior to the optic nerve, pierces the sheath of the optic nerve and runs within the nerve to the eyeball, emerging at the optic disc which enters the optic nerve
 - occlusion of this vessel or of the parent artery leads to blindness
 - The terminal branches are end arteries (arterioles), which provide the only blood supply to the internal aspect of the retina

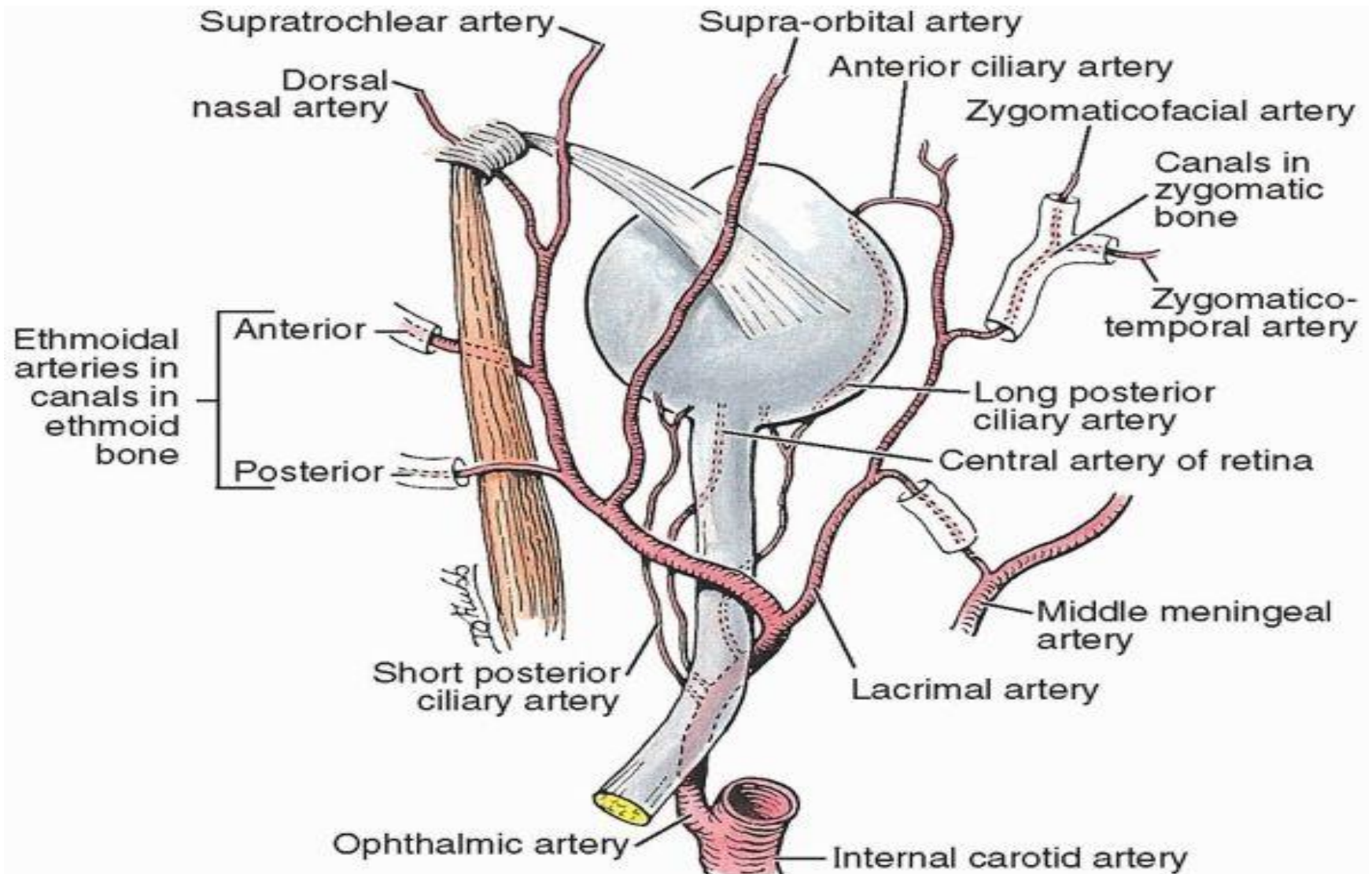
- **The long and short posterior ciliary arteries:**
 - are branches that enter the eyeball posteriorly
 - piercing the sclera, and supplying structures inside the eyeball
 - Of the eight or so posterior ciliary arteries, six short posterior ciliary arteries directly supply the choroid, which nourishes the outer non-vascular layer of the retina.
 - Two long posterior ciliary arteries, one on each side of the eyeball, pass between the sclera and the choroid to anastomose with the anterior ciliary arteries (continuations of the muscular branches of the ophthalmic artery to the rectus muscles) to supply the ciliary plexus.

- **The muscular arteries:**
 - which are branches supplying the intrinsic muscles of the eyeball
- **The supra-orbital artery:**
 - which usually arises from the ophthalmic artery immediately after it has crossed the optic nerve, proceeds anteriorly
 - exits the orbit through the supra-orbital foramen with the supra-orbital nerve
 - it supplies the forehead and scalp as it passes across these areas to the vertex of the skull

- **The posterior ethmoidal artery:**
 - exits the orbit through the posterior ethmoidal foramen to supply the ethmoidal air cells and nasal cavity
- **The anterior ethmoidal artery:**
 - which exits the orbit through the anterior ethmoidal foramen
 - enters the cranial cavity giving off the anterior meningeal branch
 - and continues into the nasal cavity supplying the septum and lateral wall, and ending as the dorsal nasal artery;
- **The medial palpebral arteries:**
 - which are small branches supplying the medial area of the upper and lower eyelids
- **The dorsal nasal artery:**
 - is one of the two terminal branches of the ophthalmic artery, leaves the orbit to supply the upper surface of the nose
- **The supratrochlear artery:**
 - is the other terminal branch of the ophthalmic artery
 - leaves the orbit with the supratrochlear nerve, supplying the forehead as it passes across it in a superior direction

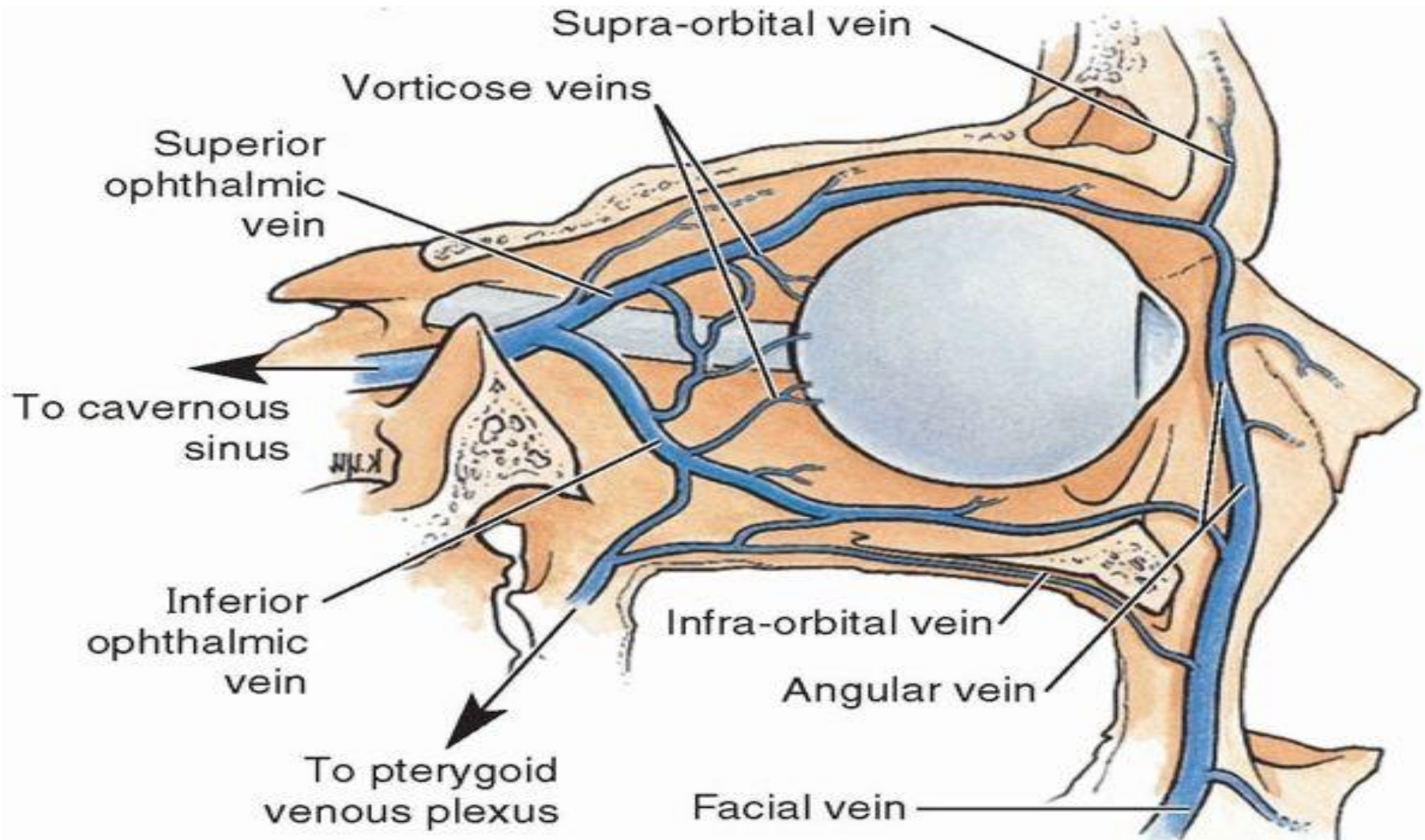
VASCULATURE OF ORBIT

- ARTERIES OF ORBIT



VEINS OF ORBIT

Venous drainage of the orbit is through the superior and inferior ophthalmic veins, which pass through the superior orbital fissure and enter the cavernous sinus

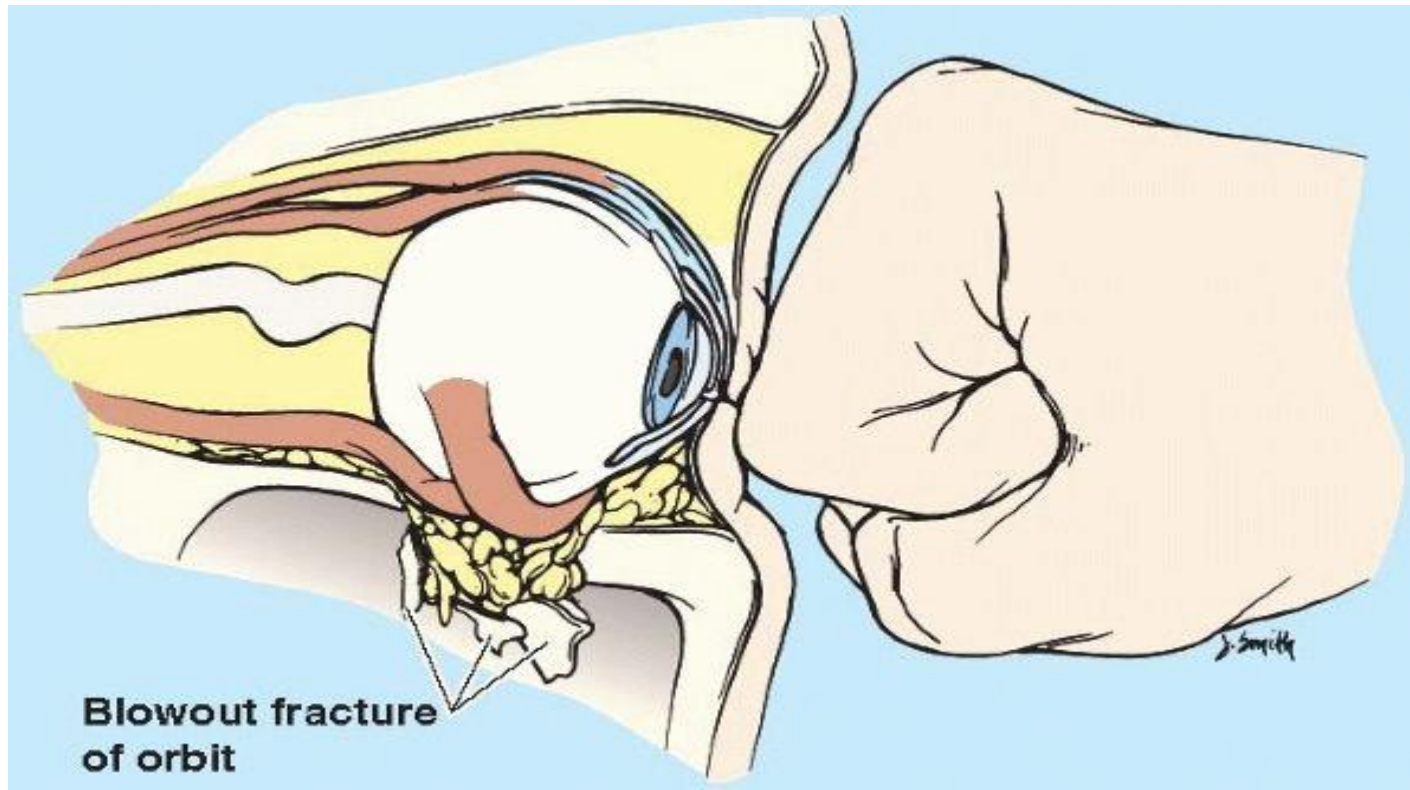


- **The central vein of the retina:**
 - usually enters the cavernous sinus directly, but it may join one of the ophthalmic veins.
 - The vortex, or **vorticosose veins**, from the vascular layer of the eyeball drain into the inferior ophthalmic vein.
 - The scleral venous sinus is a vascular structure encircling the anterior chamber of the eyeball through which the aqueous humor is returned to the blood circulation

Fractures of Orbit

- The orbital margin is strong to protect the orbital content.
- However, when the blows are powerful enough and the impact is directly on the bony rim, the resulting fractures usually occur at the three sutures between the bones forming the orbital margin.
- Because of the thinness of the medial and inferior walls of the orbit, a blow to the eye may fracture the orbital walls while the margin remains intact
- Indirect traumatic injury that displaces the orbital walls is called a “blowout” fracture.
- Fractures of the medial wall may involve the ethmoidal and sphenoidal sinuses, whereas fractures of the inferior wall (orbital floor) may involve the maxillary sinus

- Orbital fractures often result in intraorbital bleeding, which exerts pressure on the eyeball, causing **exophthalmos** (protrusion of the eyeball).
- Any trauma to the eye may affect adjacent structures—for example, bleeding into the maxillary sinus, displacement of maxillary teeth, and fracture of nasal bones resulting in hemorrhage, airway obstruction, and infection that could spread to the cavernous sinus through the ophthalmic vein



Injury to Nerves Supplying Eyelids

- a lesion of the oculomotor nerve causes paralysis of the muscle, and the superior **eyelid droops (ptosis)**.
- Damage to the facial nerve involves paralysis of the orbicularis oculi, preventing the eyelids from closing fully.
- Normal rapid protective blinking of the eye is also lost.
- The loss of tonus of the muscle in the inferior eyelid causes the lid to fall away (evert) from the surface of the eyeball, leading to drying of the cornea.
- This leaves the eyeball unprotected from dust and small particles.

Horner Syndrome

- Horner syndrome results from interruption of a cervical sympathetic trunk and is manifest by the absence of sympathetically stimulated functions on the ipsilateral side of the head.
- **The syndrome includes the following signs:**
 - constriction of the pupil (miosis)
 - drooping of the superior eyelid (ptosis)
 - redness and increased temperature of the skin (vasodilation)
 - and absence of sweating (anhidrosis).
- Constriction of the pupil occurs because the parasympathetically stimulated sphincter of the pupil is unopposed.
- **The ptosis** is a consequence of the paralysis of the smooth muscle fibers interdigitated with the aponeurosis of the levator palpebrae superioris that collectively constitute the superior tarsal muscle, supplied by sympathetic fibers

- **Paralysis of Extraocular Muscles/Palsies of Orbital Nerves**

- One or more extraocular muscles may be paralyzed by disease in the brainstem or by a head injury, resulting in diplopia (double vision).
- Paralysis of a muscle is apparent by the limitation of movement of the eyeball in the field of action of the muscle and by the production of two images when one attempts to use the muscle

OCULOMOTOR NERVE PALSY

- affects most of the ocular muscles, the levator palpebrae superioris, and the sphincter pupillae.
- The superior eyelid droops and cannot be raised voluntarily because of the unopposed activity of the orbicularis oculi (supplied by the facial nerve)
- The pupil is also fully dilated and non-reactive because of the unopposed dilator pupillae.
- The pupil is fully abducted and depressed (“down and out”) because of the unopposed activity of the lateral rectus and superior oblique, respectively.



(A) Oculomotor paralysis

ABDUCENT NERVE PALSY

- When the abducent nerve (CN VI) supplying only the lateral rectus is paralyzed, the individual cannot abduct the pupil on the affected side
- The pupil is fully adducted by the unopposed pull of the medial rectus



Abducent paralysis

Blockage of Central Artery of Retina

- Because terminal branches of the central artery of the retina are end arteries, obstruction of them by an embolus results in instant and total blindness.
- Blockage of the artery is usually unilateral and occurs in older people.

Blockage of Central Vein of Retina

- Because the central vein of the retina **enters the cavernous sinus**, thrombophlebitis of this sinus may result in the passage of a thrombus to the central retinal vein and produce a blockage in one of the small retinal veins
- Occlusion of a branch of the central vein of the retina usually results in slow, painless loss of vision

Thanks